
SENPAI: Self-Experimentation for Physical AI

An Observability-Based Research Harness

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Abstract

SENPAI (Self-Experimentation for Physical AI) is an observability-first research harness in which multi-agent state is grounded in pull requests and structured experiment logs rather than agent memory and scratchpads. Although task-agnostic, we evaluate SENPAI on CFD-surrogate recipe search, where performance depends on architecture, optimization, losses, normalization, and physics-aware considerations. SENPAI uses a thin Advisor/Student loop to turn hypotheses into PRs, training runs, and PR comments, producing an experiment ledger queryable by both agents and researchers. This makes experiments auditable, recoverable, and human-steerable. We evaluate Transolver-based SENPAI across DrivAerML, AirfRANS, and Tandem-FoilSet. Starting from the original Transolver model, SENPAI improves DrivAerML surface-pressure and wall-shear-stress relative- L_2 error, reduces AirfRANS surface-MSE below all compared references, and reaches lower Tandem-FoilSet cruise-random-uniform normalized full-field MSE than the reported benchmark. Overall, SENPAI supports sparsely steered, multi-day semi-autonomous research that can deliver strong task-specific models. We will release the harness and experiment ledger, including hosted experiment logs, so the research programme can be audited end to end.

1. Introduction

CFD surrogates offer a way to accelerate engineering design: learned models can screen early-stage geometries, explore operating regimes, and focus expensive CFD or physical testing on the most promising candidates. Recent surrogate models now handle larger meshes, irregular geome-

tries, stricter accuracy targets, and smaller training sets than were feasible only a few years ago (Zhou et al., 2026), making hybrid ML-CFD workflows increasingly attractive.

In practice, however, training a useful CFD surrogate is still a niche capability. Performance depends on architecture, optimization, normalization, boundary conditions, symmetries, and rollout stability (Wang et al., 2024). Strong recipes require both ML expertise and domain knowledge in aerodynamics or fluid dynamics (Hodges, 2024), which many engineering teams do not have. SENPAI targets this gap by exploring, documenting, and refining problem-specific surrogate recipes while keeping researchers in the loop.

Recent autonomous-research systems show that LLM agents can search literature, propose hypotheses, edit code, run experiments, and produce reports (Lu et al., 2024; Schmidgall et al., 2025; Yamada et al., 2025; Miyai et al., 2026). SENPAI builds on this direction but focuses on a different systems question: how should long-running, multi-experiment ML research be represented so that both agents and researchers can audit, recover, compare, and steer it?

SENPAI’s central design choice is to ground the research loop in external systems of record that ML teams already inspect: pull requests and git history for hypotheses, code changes, review, and provenance (GitHub Docs, 2026; Chacon & Straub, 2014); and a hosted experiment tracker for configurations, metrics, checkpoints, system telemetry, and agent traces (Biewald, 2020; Weights & Biases, 2026). Our implementation uses GitHub and Weights & Biases,¹ but the harness design is not specific to either service. Each hypothesis, code diff, training result, review decision, failed run, and baseline update becomes both machine-queryable and research-reviewable through these existing tools.

The harness loop is deliberately thin. An Advisor reads the current experiment ledger, proposes a literature-grounded hypothesis as a draft PR, and assigns it to a GPU-backed Student. The Student checks out the branch, modifies the training code, runs the experiment, logs metrics to the experiment tracker, and reports results in the PR. The Advisor

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¹Weights & Biases was the hosted experiment tracker used in the reported experiments.

then merges, requests revision, or closes the line of inquiry. Humans can steer the Advisor or Student through GitHub issues and PR comments, while markdown scratchpads remain secondary sources of state rather than the primary system of record.

Across the research programme, SENPAI produced over 3,000 PRs, 11,000 experiment-tracker runs, and operated with up to 59 concurrent Student agents, turning a multi-week semi-autonomous research programme into a durable experiment ledger.

Contributions.

- **Observable research state.** We introduce SENPAI, a semi-autonomous ML research harness that grounds experiment state in PRs, commits, and experiment-tracker runs, producing a ledger that is queryable by agents and inspectable by researchers.
- **CFD recipe search.** Starting from Transolver-style baselines, SENPAI discovers improved recipes across AirFRANS, DrivAerML, and TandemFoilSet, with strongest gains on surface and wall-shear quantities.
- **Failure modes at scale.** We analyze a 24-hour fleet trace with 53,022 Claude requests and 5.24B tokens, showing that the dominant failures were monitor-driven context bloat, brittle tool interfaces, checkpoint/result-capture gaps, and state reconciliation issues rather than failures of scientific reasoning.

2. Related Work

Recent autonomous-research systems show that LLM agents can search literature, propose hypotheses, edit code, run experiments, and produce research reports. The AI Scientist generates ideas, implements experiments, writes papers, and runs automated review (Lu et al., 2024; Sakana AI, 2024), while AI Scientist-v2 extends this direction with agentic tree search and workshop-level paper generation (Yamada et al., 2025; Sakana AI, 2025). Agent Laboratory structures literature review, experimentation, and report writing around a researcher-provided idea (Schmidgall et al., 2025), and Jr. AI Scientist iteratively improves a baseline paper using modern coding agents (Miyai et al., 2026). Related systems emphasize collaboration, shared artifacts, or domain tools: ResearchAgent generates and revises ideas over scientific-literature graphs (Baek et al., 2024), AgentRxiv lets agent laboratories upload and retrieve generated reports from a shared preprint server (Schmidgall & Moor, 2025), ChemCrow and Coscientist connect LLM planners to chemistry tools and laboratory workflows (Bran et al., 2024; Boiko et al., 2023), and FunSearch couples an LLM generator to a verifiable evaluator and scored program database (Romera-Paredes et al., 2024). Similarly, in

general ML engineering, Hugging Face’s ml-intern is a recent, non-science-focused example that reviews papers, writes training code, and runs experiments (Hugging Face, 2026). These systems establish the value of tool use, execution, and persistent artifacts in autonomous research. SENPAI builds on that direction, but studies a different substrate question: how should the state of a long-running, multi-experiment ML research programme be represented so that both agents and researchers can inspect, compare, recover, and steer it?

The closest systems to SENPAI also recognize that long-horizon agents need state outside the model context, but they choose different substrates. Kosmos maintains an internal structured memory of literature-search and data-analysis summaries to choose later tasks, and cites final report claims with papers or generated Jupyter notebooks (Mitchener et al., 2025). AiScientist uses a permission-scoped File-as-Bus workspace so agents can re-ground on file-based analyses, plans, code, logs, and experimental evidence (Chen et al., 2026). Grounded autonomous research externalizes state as rerunnable simulation artifacts and comparison results within a single-paper reproduction loop (Huang, 2026). SENPAI takes a complementary systems position: the authoritative state should be the MLOps tooling machine learning researchers already use. In SENPAI, pull requests carry hypotheses, code review, discussion, and review decisions (GitHub Docs, 2026); git history provides ordered code provenance (Chacon & Straub, 2014); and hosted experiment trackers provide structured records of metrics, hyperparameters, artifacts, checkpoints, system telemetry, and traces (Biewald, 2020; Weights & Biases, 2026). This distinction matters most at scale: once an autonomous research system produces hundreds or thousands of experiments, the central challenge is no longer only whether an agent can run an effective experiment, but whether researchers can understand the programme it is creating.

3. System Design

Core principle. SENPAI’s central design principle is that authoritative experiment state lives in pull requests, git history, and experiment-tracker runs, not in agent memory or local scratchpads. The harness is therefore organized around making each hypothesis, code change, training result, and review decision durable, queryable, and inspectable through existing research tools.

PR-mediated lifecycle. SENPAI is a thin two-role harness over a task repository. An Advisor reads the current experiment ledger, optionally calls a literature-search sub-agent, opens draft PRs containing hypotheses and experiment instructions, and assigns them to GPU-backed Stu-

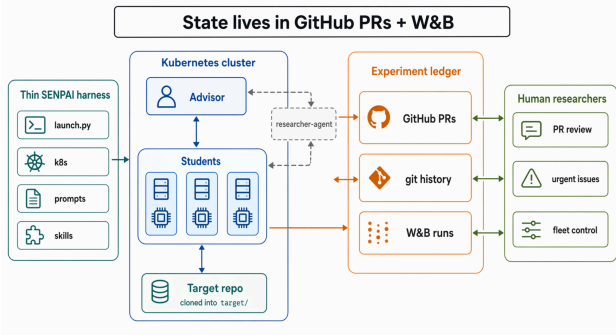


Figure 1. SENPAI deployment architecture. A thin harness deploys one Advisor and N Student pods; both roles can call a shared literature-search sub-agent. All programme state lives outside the cluster in the experiment ledger: PRs, git history, and experiment-tracker runs.

dents. Each Student claims its labeled PR, edits the branch, launches training with experiment-tracker logging, and reports results back as PR comments. Researchers can steer Advisors or Students via labeled GitHub issues or by leaving comments on a PR. A hypothesis’s trajectory therefore remains visible as a standard PR history linked to experiment-tracker runs, rather than as an agent-internal trace.

Design choices. SENPAI follows three design choices: externalizing experiment state into existing systems of record, keeping the autonomous loop thin, and exposing the same ledger to both agents and researchers. The role prompts, problem-program contract, helper skills, and outer-loop pseudocode are summarized in Appendices A to D.

3.1. Externalized Experiment State

The experiment ledger has two linked records. The PR records the hypothesis, implementation instructions, code diff, Student report, Advisor review, and final decision. The experiment tracker records the run details: configuration, metrics, checkpoints, system data, and agent traces. In our experiments, this tracker was Weights & Biases. Together these records are inspectable through familiar code-review and experiment-tracking interfaces and queryable through mature CLIs, allowing both researchers and agents to audit individual experiments or summarize programme-level progress.

The Advisor also maintains lightweight markdown summaries for focus, such as current baseline metrics, dataset notes, and open research directions. These files are treated as caches rather than authoritative state: when summaries disagree with PRs, git history, or experiment-tracker runs, the ledger wins.

3.2. Thin Advisor/Student Loop

Each experiment follows a PR-mediated lifecycle. The Advisor drafts a PR with a hypothesis, implementation guidance, target metrics, baseline values, and a Student label. The Student implements the change on the PR branch, runs training with experiment-tracker logging, and posts commands, training run links, metrics, and experiment interpretation as a PR comment. Once the PR is marked ready for review, the Advisor compares the result against the stated baseline and either merges the improvement, requests revision, or closes the branch. Merged PRs advance the programme baseline used for subsequent hypotheses.

SENPAI deploys as a small set of Kubernetes workloads: one Advisor CPU pod, N Student GPU pods, and a shared persistent data volume for datasets and checkpoints. The harness image is stable across problems; a task repository can be swapped in per SENPAI research programme, so every agent commit, branch, and PR lives in a self-contained target repository that a researcher can inspect independent of the harness.

The outer harness loop is programmatic rather than agentic. A shell wrapper polls GitHub and Student state for review-ready PRs, human issues, new comments, and idle Students; if no action is available, it sleeps without invoking the LLM. When action is required, the wrapper starts or resumes the Advisor with a triage message. This keeps routine polling outside the model context and focuses the agent on research decisions rather than heartbeat bookkeeping.

3.3. Human Steering Through Shared Observability

Researchers primarily steer SENPAI’s Advisor or Students through GitHub issues, while PR comments can also provide experiment-level intervention. This channel is intentionally low bandwidth: SENPAI is designed for sparse steering rather than continuous supervision, and each intervention remains attached to the public experiment record it affected.

In our longest observed SENPAI deployment on DriveAerML, the system ran continuously for 9 days under a 4.5-hour per-experiment budget, opening 289 PRs across an 8-student active fleet with 10 issue-based interventions from researchers.

4. Experiments and Results

4.1. Experiment Setup

Harness. Claude Code (v2.1.85 to v2.1.117) was run in headless mode as the agent harness for both the Advisor and Students. Claude Opus 4.6 and Claude Opus 4.7 were used with a compaction trigger at 200k tokens. Student pods were scheduled on NVIDIA RTX 6000 Blackwell nodes

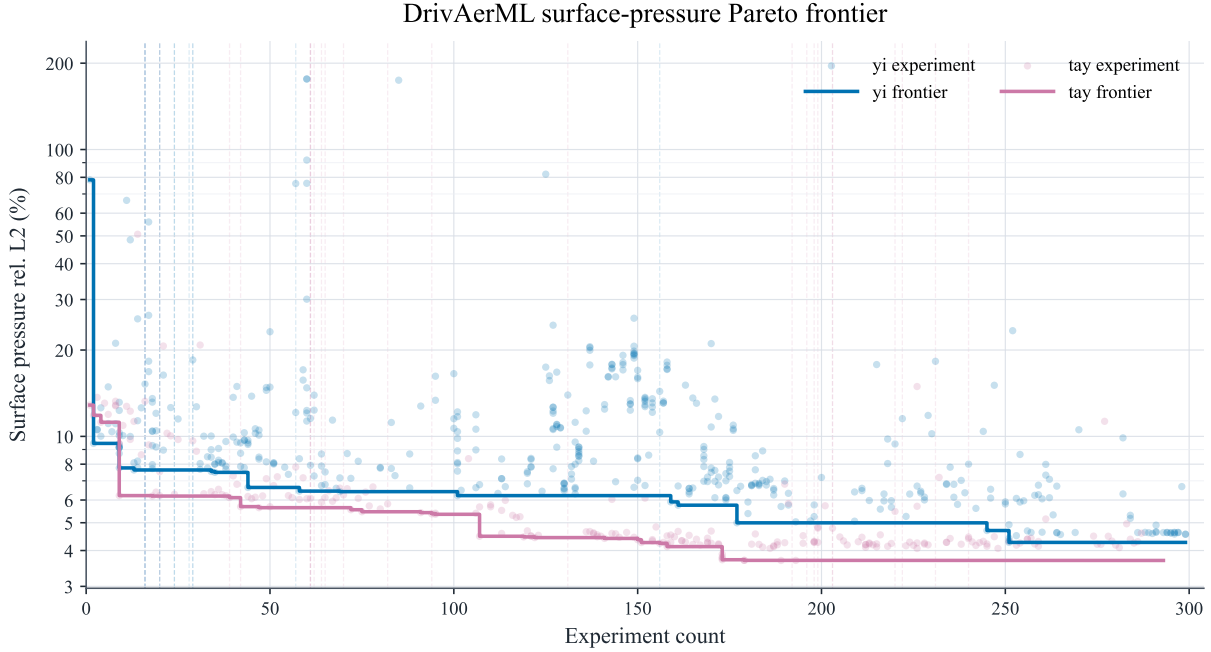


Figure 2. Evolution of DrivAerML surface-pressure test error across two SENPAI campaigns. Faded markers show individual completed experiments from the `yi` and `tay` branches, while solid curves show the cumulative Pareto frontier: the best surface-pressure relative L_2 error achieved up to each experiment count. The vertical lines mark GitHub-recorded human researcher/operator interventions. `tay` was instructed to use DDP8 and was given a 6 hour per-experiment training budget, hence the better performance.

with 96 GB VRAM per GPU, from 1 to 8 GPUs per Student node. At maximum, $N = 59$ concurrent Students were tested, with peak concurrent training runs of 347. Over the five-week experimentation and harness-development cycle, 30 billion tokens were processed by Claude Code while running physical-AI experiments.

Scale. Over the course of the research programme, SENPAI created over 3,000 PRs and recorded over 11,000 experiment-tracker runs while deploying, at peak, 59 distinct Student agents. Approximately 30 billion Claude Code tokens were processed during experimentation.

4.2. Datasets and Metrics

To demonstrate SENPAI’s CFD performance across multiple benchmarks, we train and evaluate on three CFD surrogate datasets spanning 2D tandem-airfoil flow, 2D airfoil RANS, and 3D automotive aerodynamics: TandemFoilSet (Lim et al., 2026), AirfRANS (Bonnet et al., 2022), and DrivAerML (Ashton et al., 2024).

TandemFoilSet. We use the paper’s original Experiment 4 partition, Cruise Random and Race Car tasks, as one of two TandemFoilSet datasets and measure denormalized surface-pressure MAE. In addition, we introduce a sec-

ond dataset split, TandemFoilSet-Balanced (Anonymous, 2026b), with 1,499 training cases across four balanced test splits: one single-foil in-distribution split, two unseen-front-camber geometry holdouts (race-car and cruise), and one Reynolds-number holdout. An equal-weight average denormalized surface-pressure MAE of these four splits is used as the test metric.

AirfRANS. We use the official full task with the last 10% of the official training list held out for validation, giving a 720/80/200 train/validation/test split while keeping the official test set unchanged. We test on normalized targets on the official test split.

DrivAerML. We use the public processed-case split, 400/34/50 train/validation/test, and report surface-pressure, vector wall-shear-stress, and volume-pressure relative- L_2 , computed per case on unnormalized predictions and then averaged over test cases.

4.3. DrivAerML

Result. On the DrivAerML benchmark, SENPAI, starting from a Transolver-based implementation, improves on the original Transolver’s test-set metrics of surface-pressure relative- L_2 error (3.88% versus 4.81%) and vector wall-

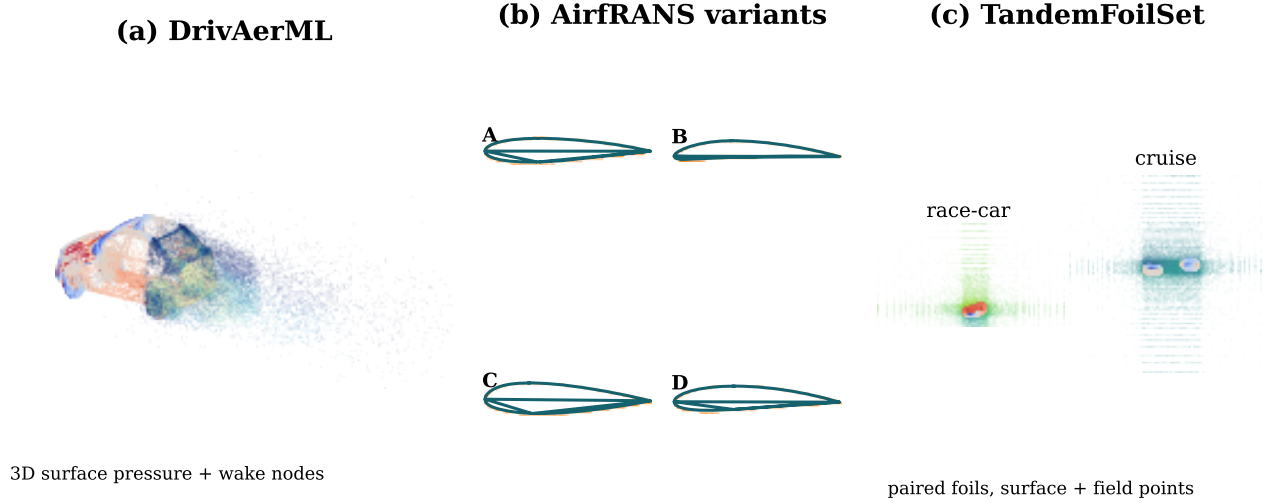


Figure 3. CFD surrogate problems used by SENPAI, shown as geometry-and-field examples rather than benchmark results. (a) DrivAerML is a 3D automotive aerodynamics task with millions of surface and volume sample points; the panel shows pressure-colored surface samples together with downstream wake nodes. (b) AirfRANS contains families of 2D airfoil cross-sections. (c) TandemFoilSet contains paired-foil fields, here shown for race-car and cruise configurations with surface nodes overlaid on field samples.

Table 1. DrivAerML test relative- L_2 error (%). Lower is better.

Method	p_s	τ	p_v
SENPAI, ensemble	3.69	6.86	11.36
Transolver-3 (Zhou et al., 2026)	3.71	5.85	5.72
AB-UPT (Alkin et al., 2025)	3.82	7.29	6.08
SENPAI, single model	3.88	7.06	10.76
Transolver++	4.12	6.42	6.70
Transolver (Wu et al., 2024)	4.81	8.95	7.74

shear-stress relative- L_2 error (7.06% versus 8.95%). The selected single model also improves on AB-UPT for wall shear (7.06% versus 7.29%), while volume pressure remains behind public references (10.76% versus 6.08% for AB-UPT and 5.72% for Transolver-3). Figure 2 shows the semi-autonomous experiment trajectory for surface-pressure test error across earlier SENPAI campaigns.

Recipe. The selected single-model recipe SENPAI found used the original Transolver backbone but scaled it to 4 layers, 512 hidden units, and 128 slices, replaced fixed sinusoidal coordinates with a separable learnable Fourier positional embedding initialized across multiple spatial scales (Li et al., 2021; Schenck et al., 2025), and trained with Lion (Chen et al., 2023), EMA, and GradNorm adaptive loss balancing (Chen et al., 2018). It trained with 8-GPU DDP for an effective batch size of 16 under a 50-epoch budget, with the best EMA checkpoint selected at epoch 27. A notable empirical finding from SENPAI was the application of this lightweight STRING-inspired encoding from computer vision and robotics use cases to CFD surrogate modelling. Full run IDs and hyperparameters for the selected

single model and ensemble are listed in Appendix E.

Caveat. Finally, a greedy inference-time ensemble found by SENPAI improved on all reported surface-pressure relative- L_2 metrics for DrivAerML (3.69%) and improved on AB-UPT for wall-shear relative- L_2 error (6.86%). A large gap in test-set volume-pressure relative- L_2 error remains for the best single model (10.76%) relative to AB-UPT and Transolver-3, potentially in part due to the shorter training time of SENPAI compared to the 500-epoch trainings of AB-UPT and Transolver-3. This training-duration discrepancy highlights an important decision criterion in autonomous research training: whether to scale up to longer, more expensive training runs after observing promising smaller-scale experimental results.

4.4. AirfRANS

Result. On the AirfRANS full benchmark, SENPAI, starting from a Transolver-based implementation, improves substantially on the original Transolver surface-MSE result (5-seed mean 0.0013 versus 0.0142) and on the strongest published surface reference, SpiderSolver (0.0043). The same model obtains mean volume-MSE of 0.00451, improving on GeoANF but remaining above SpiderSolver and slightly above the original Transolver volume result. SENPAI therefore achieves the strongest reported surface-MSE among these references, while the full benchmark remains limited by volume accuracy.

Recipe and caveat. The final AirfRANS recipe SENPAI found used a 2-layer, 256-hidden-unit, 4-head SENPAI-

Table 2. AirfRANS full-task test MSE. Targets are train-statistic normalized, per-case means on the official test split; lower is better. Spider denotes SpiderSolver (Qi et al., 2025).

Metric	SENPAI	Spider	GeoANF	Transolver
Surface MSE	0.00130	0.0043	0.0089	0.0142
Volume MSE	0.00451	0.0017	0.0062	0.0037

Table 3. TandemFoilSet-Balanced 5-seed mean denormalized surface-pressure MAE. Lower is better.

Metric	MAE
Single-foil in-distribution	17.2905
Race-car camber holdout	31.2177
Cruise camber holdout	28.3145
Reynolds-number holdout	16.9834
Average	23.4515

Transolver with 96 slices, fixed multiscale Fourier coordinate features, a zero-initialized surface refinement head, Lookahead-wrapped AdamW, cosine scheduling, gradient clipping, weight decay, EMA with target decay 0.999, and a volume-weighted normalized-MSE objective with volume loss weight 10.0. All five seeds beat the published SpiderSolver surface-MSE result, while volume-MSE remains above SpiderSolver and more sensitive to seed. The complete five-seed configuration is reported in Appendix F.

4.5. TandemFoilSet

Result. Under the TandemFoilSet Experiment-4 task, SENPAI found a Transolver-based improvement on the cruise random uniform full-field MSE of the original paper ($1.7 \cdot 10^{-3}$ versus $1.0 \cdot 10^{-1}$). On TandemFoilSet-Balanced, SENPAI found a recipe that produced a denormalized surface-pressure MAE of 23.45 using a 5-seed average.

Recipe signal. The PR ledger shows the path: lower learning rates near $1e-4$, gradient clipping, EMA, warmup, and schedule-length tuning.

4.6. Flat-Agent Comparison

Setup. To contrast SENPAI’s Advisor-mediated loop, we compared the performance of a *kagent* harness (Anonymous, 2026a) against the SENPAI harness using the TandemFoilSet-Balanced dataset and the same compute budget. *kagent* is a lighter-weight harness that drops the Advisor/Student split in favor of a flat Kaggle-like competition with a shared leaderboard; each agent keeps its own experiment journal and is prompted to compete aggressively to climb the leaderboard. In contrast to a Kaggle competition, the agents also have access to the same hosted experiment-logging project and commit history, so they can

Table 4. TandemFoilSet-Balanced 12-hour harness comparison. Lower average surface-pressure MAE is better.

Metric	SENPAI	<i>kagent</i>
Best avg-surf- <i>p</i> MAE	44.15	34.58

learn from other agents’ work. The *kagent* prompt and the full comparator run summary are provided in Appendix G.

Interpretation. In a direct 12-hour head-to-head, *kagent* behaved like a public Kaggle sprint: agents trained short variants, watched the leaderboard and shared hosted-logger configs, and quickly reused recipes that worked for peers. This fast peer diffusion, including copied hyperparameters and occasional prediction ensembles, likely gave it an exploitation-speed edge over SENPAI’s slower PR-mediated review loop. While this was successful in a 12-hour experiment setting, the inability to steer these agents and the less rigorous polling and monitoring infrastructure are likely to lead to drift and context rot as each Claude Code agent’s context window fills.

5. Discussion

5.1. Observable State as a Control Surface

Control surface. The PR and experiment-tracker ledger made SENPAI useful as a research assistant rather than a tool to execute experiments. Because hypotheses, code diffs, metrics, failures, and review decisions were all recorded in systems researchers already use, humans could inspect the research programme without needing to read the agents’ context. The same records also gave agents a stable system of record: they could query prior runs, identify stale experiments, compare seed behavior, and decide whether a new result actually advanced the baseline. In SENPAI, observability is therefore not simply logging; it is the layer through which both humans and agents can steer the research programme.

Recovery. This externalized state also reduced the fragility of long-running agent sessions. When agents compacted, restarted, or lost local scratchpad context, the next cycle could reconstruct the experiment from the PR description, code diff, git history, experiment-tracker records, and posted result comments. Local summaries remained useful for focus, but they were treated as caches over a more durable experiment record.

Research programme. The ledger also turned a collection of individual experiments into a queryable research programme. Researchers and agents could ask what had been tried, which failures were informative, which recipes transferred across datasets, where progress had stalled, and

which resources were idle. This is SENPAI’s central systems claim: the goal is not to replace the researcher, but to increase the bandwidth at which researchers can understand, guide, and accelerate experimentation.

5.2. Harness Tradeoffs

Tradeoff. The kagent comparison highlights a harness-design tradeoff. Flat, leaderboard-driven cohorts can share successful recipes quickly because agents observe peer results directly and optimize for short-horizon score improvement. In contrast, SENPAI’s PR-mediated loop is slower but preserves review decisions, experiment provenance, and sparse human steering. We view kagent as useful for rapid exploitation, and SENPAI as better suited to longer-running programmes where auditability and recovery matter alongside score.

5.3. Failure Modes: Where Long-Running Research Agents Break

Observed failures. SENPAI’s most common failures were harness-engineering failures: monitoring long-running jobs, recovering across compaction or restarts, enforcing brittle tool interfaces, and reconciling PR state with experiment-tracker state. They were less often failures of scientific reasoning. This distinction matters because it suggests that improving autonomous research systems requires better observable infrastructure, not only stronger prompts or more capable models. Additional artifact schemas and the extended taxonomy are listed in Appendix H.

Monitor bloat. During development, one dominant cost to running SENPAI was monitor-driven context bloat. In one notable experiment, Students monitored training logs for progress, errors, checkpoints, and completion, but persistent `tail -f | grep` events repeatedly resumed long-lived Student sessions for low-information log lines. In a 24-hour window with 57 Students running, 615 monitor setups produced 28,247 agent-facing monitor events and 28,246 agent responses, consuming 3.25B cache-inclusive tokens. The median agent response reloaded roughly 110k cached tokens while adding only 347 new uncached input tokens. This is an example of an architectural failure mode when running long-running research systems: training monitors should be cheap and escalation-based, surfacing only relevant events such as errors, experiment completion, or metric milestones.

Tool interfaces. Tool-use errors were smaller but systematic: in the same 24-hour window, 892 of 25,365 tool results had errors. The main causes were failed shell commands, GitHub scope errors, blocked wait patterns, wrong paths, failed pushes, oversized reads, and edit-guard

failures. These errors point to a practical lesson: high-frequency harness operations should be executable helpers with narrow interfaces, rather than repeatedly reconstructed shell or GitHub operations inside agent context.

State drift. Compaction introduced a separate risk. The 24-hour window contained 240 automatic compactions across the 57 Students, with Student contexts reduced from roughly 196k tokens to 3k-token summaries. These summaries were sufficient for resumption, but lossy: state drift appeared in the agent conversation logs with mistakes such as incorrect GitHub labels or confusion over live experiment state. SENPAI’s experiment ledger mitigated this risk by externalizing much of the key experiment state. As a result, in that 24-hour window, despite an average of 4.2 compactions per Student, 38 of 39 Student sessions whose monitors showed training had completed also left PR comments as expected, and 33 of 39 PRs reached ready/status-review handoff as expected, indicating that despite multiple compactions, Students were able to use the experiment ledger’s external state to course-correct and take experiments to completion.

Mitigation. From observing these failure modes, we conclude that long-running research agents need robust, observable infrastructure more than additional prompt prose. Monitor bloat requires thoughtful observation; tool errors require executable helpers; training failures require mandatory result and checkpoint capture; state drift requires a single authoritative ledger, with summaries treated as caches. SENPAI did not eliminate failure, but it made failure measurable, attributable, and repairable.

5.4. Limitations

Scope. SENPAI is task-agnostic as a harness, but our empirical evidence comes from CFD-surrogate recipe search. We have not yet evaluated whether the same design transfers to domains with different data modalities, evaluation loops, or laboratory constraints.

Surface metrics. The strongest empirical results are on surface quantities. This matches many aerodynamic design workflows, where surface pressure and forces are primary signals, but volume-field prediction remains less developed; for example, our AirfRANS and DrivAerML experiments improve surface MSE while remaining behind the strongest published methods on volume MSE.

Supervision. SENPAI is semi-autonomous rather than fully autonomous. It requires an initial task repository, metric definition, dataset setup, and occasional human steering. The GitHub Issues and PR-comment interface is auditable and low bandwidth, but slower than interactive supervision.

Infrastructure dependence. Finally, SENPAI depends on mature external systems of record and substantial compute/API budget. In our implementation these were GitHub and Weights & Biases, which improved observability but also introduced dependence on hosted-service reliability, access control, rate limits, and retention policies. We will release the harness and experiment ledger so future work can audit costs, reproduce results, and evaluate more efficient variants.

6. Conclusion

SENPAT shows that a lightweight, observability-first harness can sustain semi-autonomous ML research when multi-agent state is grounded in the artifacts researchers already inspect: pull requests, git history, and experiment-tracker runs. Across a multi-week CFD-surrogate research programme, this design supported thousands of autonomous experiments over three benchmarks while allowing researchers to steer the system through ordinary PR review and a GitHub Issues channel.

The main lesson is that reliability comes from the harness, not the agent instructions alone. Long-running research agents did not primarily fail because they lacked clear instructions; they failed at system boundaries: brittle tool interfaces, unbounded monitoring, missing checkpoint capture, and drift between derived summaries and the authoritative ledger. Prompt hardening alone did not remove these failure modes. Durable external state, clear scripts for passing work between agents, filtered monitoring, and mandatory result capture were more important for keeping the research programme recoverable and auditable.

SENPAT’s contribution is therefore not to replace the researcher, but to increase the bandwidth at which researchers can understand, guide, and accelerate autonomous experimentation. By making both the inner experiment loop and the outer harness loop observable through mature systems of record, SENPAT turns autonomous research from an opaque agent workflow into a queryable, reviewable research programme. We will release the harness, experiment ledger, and failure-mode analysis to support future work on more reliable and efficient autonomous research systems.

Software and Data

For double-blind review, non-anonymous links to the harness repository and hosted run ledger are withheld from this version and will be included in the camera-ready version. The CFD datasets used in the experiments are DrivAerML, AirfRANS, and TandemFoilSet; access and redistribution follow the licenses and terms provided by the original dataset authors.

Impact Statement

This work aims to make autonomous ML research more auditable, recoverable, and steerable by grounding agent activity in systems of record that researchers already inspect. The positive impact is increased researcher bandwidth for expensive scientific ML experimentation, especially in domains where strong baselines require repeated empirical iteration. The main risks are increased compute/API consumption, over-trusting autonomous experiment choices, and propagating errors when agent-written summaries drift from measured evidence. SENPAT mitigates these risks by keeping humans in the review loop, preserving experiment provenance in PRs and tracked runs, and treating summaries as caches rather than authoritative state.

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A. Agent Instructions and Source Map

The experiments in this paper were controlled by a small number of versioned text artifacts rather than by hidden agent memory. The exact prompt files will be released with the SENPAI harness; this appendix summarizes their operational content, and Appendix I includes the full text of the source prompts and programme contracts.

Table 5. Primary instruction artifacts used by the harness and comparator.

Artifact	Location	Role in the system
Advisor prompt	<code>system_instructions/CLAUDE-ADVISOR.md</code>	Defines the no-GPU research lead: survey PRs, review result comments and W&B runs, merge validated winners, close dead ends, and assign one-hypothesis PRs to idle Students.
Student prompt	<code>system_instructions/CLAUDE-STUDENT.md</code>	Defines the GPU worker: claim only assigned PRs, edit the task training file, run bounded experiments, report exact commands, metrics, W&B IDs, and structured terminal results.
Researcher sub-agent	<code>.claude/agents/researcher-agent.md</code>	Defines a literature and experiment-history specialist used to produce mechanism-grounded experiment proposals and research-state updates.
Problem contract	<code>DrivAerML/program.md</code>	Example task contract specifying mission, data, editable files, metric keys, checkpoint selection, telemetry, and paper-facing target values.
TandemFoil2 contract	<code>target/tandemfoil2/program.md</code>	Active local problem contract for the kagent-seeded TandemFoil2 package, including split names, edit boundaries, and the <code>val/12_error</code> leaderboard metric.
Flat comparator prompt	<code>KAGGLER_AGENT.md</code>	Defines the kagent loop: watch leaderboard and W&B, edit local training/prediction code, keep only useful code changes, always journal outcomes, and continue until interrupted.

The most important verbatim prompt element is the terminal result marker, which makes a Student’s PR comment machine-checkable:

```
SENPAI-RESULT: {"terminal":true,"status":"complete","pending_arms":false,
"wandb_run_ids":["<run-id>"],"primary_metric":{"name":"<metric>","value":<number>},
"test_metric":{"name":"<metric>","value":<number>}}
```

The Advisor prompt additionally contains a plateau protocol: after five or more non-improving experiments, the Advisor must escalate from local tuning toward a different strategy tier, revisit first principles, use the researcher sub-agent, and propose larger mechanism changes rather than treating the plateau as a stopping signal.

B. Problem-Program Contract

Each SENPAI target repository carries a `program.md` file that makes task assumptions explicit. The DrivAerML contract used as a reference artifact defines the elements in Table 7; the local TandemFoil2 contract follows the same pattern, with `val/12_error` averaged over four validation splits as its leaderboard metric. These files are intentionally more specific than generic prompts: they are the task-local source of truth for what a valid experiment may edit, how data are structured, and which metrics count as paper-facing evidence.

C. Skill Catalogue

SENPAI’s prompts route common operations through small helper skills and a shared `senpai-gh` library. This was a deliberate control choice: high-frequency GitHub and bookkeeping operations should have narrow, testable interfaces instead of being reconstructed from scratch by each agent invocation.

D. Advisor Outer Loop

Table 6. Role contracts induced by the prompts.

Role	Inputs	Allowed actions	Required output
Advisor	PR ledger, W&B runs, Student pod state, human GitHub issues, and <code>program.md</code> .	Create draft PRs, assign labels, review each completed PR, request revisions, close dead ends, and merge winners after structured terminal results.	Updated baseline and research-state files, PR review comments, and new hypothesis PRs for idle Students.
Student	Assigned PR, target branch, <code>program.md</code> , training logs, W&B runs, and Advisor comments.	Modify the task training file, run the assigned experiment, debug crashes/OOMs, and resubmit after requested changes.	A PR comment beginning with a single-line SENPAI-RESULT JSON marker, followed by metrics, run IDs, commands, interpretation, and follow-ups.
Researcher	Prior PRs, W&B evidence, code lineage, papers, repos, and domain references.	Search literature and experiment history, reconstruct mechanisms, and design discriminating follow-up experiments.	A decision-useful synthesis: mechanism, evidence, implementation notes, proposed experiment tree, stop condition, and confidence.
kagent	Shared leaderboard, own experiment journal, W&B configs, visible commits, and local logs.	Iterate directly on <code>train.py</code> and <code>predict.py</code> , run training/prediction, retain or discard code, and write the journal after every trial.	Scored submissions plus a persistent experiment journal, with no Advisor review gate.

The key property is that ordinary polling remains outside the LLM. The Advisor is invoked only when there is actionable state: a PR to review, a human issue to answer, or idle Students that need assignments. The watermark advances only after successful invocations, so transient failures retry against the same triage state.

E. DrivAerML Recipe Details

The `string_multisigma` encoding is a lightweight adaptation of separable translationally invariant position encodings (Schenck et al., 2025): per-axis learnable sine/cosine bases are initialized across several spatial scales and projected into the Transolver width.

The ensemble result in Table 1 is documented in PR #602. It was a greedy forward-selection ensemble over test predictions from seven independently trained checkpoints: `nh96x7m4`, `5o7jc7wi`, `wyz68o8r`, `9mm3sz7x`, `49aimdiz`, `19qf6di1`, and `nh2ke150`. Candidates were added one at a time to minimize validation field error, and the final inference-time prediction averaged the selected members uniformly.

F. AirfRANS Recipe Details

G. kagent Comparator Details

The kagent comparator used a flat competition prompt rather than SENPAI’s Advisor/Student split. Its prompt instructs each agent to read the task README and experiment journal before coding, check the leaderboard and W&B before each iteration, formulate a hypothesis, modify only the training/prediction entypoints, train and predict, keep code only when it improves, and always append an experiment-journal entry even for failed or discarded trials. The primary metric was validation L2 error, with memory pressure, turbulence, and no-slip boundary conditions called out as domain challenges. Unlike SENPAI, this loop is explicitly autonomous once started and does not pause for Advisor review.

TandemFoilSet parity run. An eight-agent kagent cohort ran on the TandemFoilSet parity benchmark from 2026-04-23 15:53 UTC to 2026-04-24 04:00 UTC, using one GPU pod per agent, a 30-minute per-training-run budget, and the Transolver starter. The run produced 330 agent commits and 223 scored leaderboard updates without human intervention. Agents communicated only through shared data/prediction volumes, the git remote and leaderboard, and W&B telemetry.

The most important qualitative finding was rapid peer diffusion. Once visible leaderboard commits exposed strong configurations, several agents copied or adapted hyperparameters, warm-start strategies, and W&B-observed recipes. The decisive improvement was a full-mesh fine-tuning unlock: several top agents concluded that training on heavily subsampled point

Table 7. Information supplied by a task `program.md`.

Contract component	Example content in the DrivAerML programme
Mission and targets	Minimize held-out <code>test_primary/*</code> metrics for surface pressure, vector wall shear, and volume pressure, using validation for steering but final test metrics for claims.
File boundaries	<code>train.py</code> , <code>model.py</code> , and runtime helpers are experiment-editable; data loaders, manifests, and split files are read-only during normal experiment PRs.
Data contract	Surface tensors include coordinates, normals, area, pressure coefficient, and wall-shear components; volume tensors include coordinates, signed-distance field, and volume pressure.
Split and evaluation	The public processed split is fixed at 400 train, 34 validation, and 50 test cases, with deterministic full-fidelity evaluation chunks for validation and test.
Model interface	The reference model consumes padded surface and volume tensors plus masks, and must not compute losses or metrics on padding tokens.
Telemetry and validity	W&B must contain finite final primary metrics, plus gradient, weight, slope, validation, and test diagnostics sufficient to diagnose unstable or invalid runs.
Paper-facing metrics	Individual target-family relative- L_2 percentages are reported; the aggregate checkpoint metric is treated as a triage scalar rather than a benchmark column.
Workflow	Roles are coordinated through PRs and GitHub issues; final result claims must be traceable to run IDs and code state.

Table 8. Helper skills used by the Advisor and Students.

Helper	Caller	Effect on the experiment ledger
<code>survey-prs</code>	Advisor	Reads review-ready, WIP, draft/stalled, and idle-Student state for an Advisor branch.
<code>poll-for-work</code>	Student	Returns assigned WIP PRs for a Student label and reads comments to detect revisions.
<code>check-human-issues</code>	Both	Finds labeled human issues for the caller and posts role-prefixed replies without closing issues.
<code>list-experiments</code>	Researcher	Exports merged winners, compact result tables, and detailed PR bodies from the PR ledger.
<code>assign-experiment</code>	Advisor	Creates a Student branch, opens a draft PR, labels it <code>status:wip</code> , and writes full experiment instructions.
<code>submit-experiment-results</code>	Student	Commits changed experiment files, pushes the branch, marks the PR ready, and swaps <code>status:wip</code> to <code>status:review</code> .
<code>merge-winner</code>	Advisor	Preflights terminal results, squash-merges a winning PR, updates the Advisor baseline, and pulls the updated branch.
<code>senpai-gh</code>	Both	Provides lower-level label swaps, PR comments, closure, preflight, and read-only query primitives.

sets created a train/eval mismatch for attention layers that pool over nodes, and moved from subsampled pretraining to batch-size-2 full-mesh fine-tuning. Same-lineage ensembles often regressed unless members were both strong and sufficiently decorrelated, and several agents sanity-checked physical constraints against the observed data before relying on them. A separate scorer-race incident marked partially written prediction directories as terminally incomplete, illustrating the same infrastructure-contract fragility discussed in Appendix H.

External competition run. As an additional stress test, sixteen Claude Opus 4.6 kagent agents were pointed at the GRaM ICLR 2026 open competition from 2026-03-29 to 2026-03-30. The winning cohort member reached 0.8526 mean L_2 against a roughly 1.75 reference-MLP baseline, and the strongest checkpoint was packaged and submitted to the upstream competition as a live pull request. This run predates the later experiment-journal convention, so its evidence is primarily leaderboard and commit based.

H. Evidence Artifacts and Failure Taxonomy

The experiment ledger supports several supplementary audit views beyond the main tables:

- full benchmark result tables for AirfRANS, TandemFoilSet parity, TandemFoilSet-Balanced, and DrivAerML validation/test trajectories;

Algorithm 1 Advisor triage loop outside the LLM context

```

1:  $iteration \leftarrow 0$ ;  $last\_check \leftarrow \emptyset$ 
2: while true do
3:    $iteration \leftarrow iteration + 1$ 
4:    $review\_ready \leftarrow \text{LISTREVIEWREADYPRs}(branch, last\_check)$ 
5:    $human\_issues \leftarrow \text{LISTNEWHUMANISSUES}(branch, last\_check)$ 
6:    $idle\_students \leftarrow \text{LISTIDLESTUDENTS}(students, branch)$ 
7:   if  $iteration > 1$  and  $review\_ready$ ,  $human\_issues$ , and  $idle\_students$  are empty then
8:     sleep for one harness tick and continue
9:   end if
10:   $message \leftarrow \text{RENDERTRIAGE}(review\_ready, human\_issues, idle\_students)$ 
11:  if  $iteration = 1$  then
12:     $exit\_code \leftarrow \text{RUNAGENT}(role\_prompt + message)$ 
13:  else
14:     $exit\_code \leftarrow \text{RESUMEAGENT}(heartbeat + message)$ 
15:  end if
16:  if  $exit\_code = 0$  then
17:     $last\_check \leftarrow \text{MAXUPDATEDAT}(review\_ready, human\_issues)$ 
18:  end if
19:  sleep for one harness tick
20: end while

```

Table 9. Selected DrivAerML SENPAI single-model configuration.

Setting	Value
Source	PR #740 on driværml-long-20260504; merge commit 9130201
W&B run	5x8wofzm
Training budget	50-epoch budget; best EMA checkpoint at epoch 27
Parallelism and batch	8-GPU DDP; batch size 2 per rank, 16 global
Optimizer	Lion, learning rate 1e-4, weight decay 1e-4
EMA	0.999
Backbone	4 layers, 512 hidden units, 4 attention heads, 128 slices
Positional encoding	string_multisigma; sigmas 0.25, 0.5, 1.0, 2.0, 4.0
Point sampling	Train: 40,000 surface / 65,000 volume; eval: 40,000 surface / 65,000 volume
Augmentation	None in selected run
Loss	Normalized MSE with GradNorm ($\alpha = 0.5$, learning rate 1e-3) and base surface/volume weights 1.0/1.0

- an experiment-ledger CSV schema with PR number, title, status, Student, W&B run IDs, primary metric name/value, baseline delta, era tag, and merge-decision timestamp;
- redacted examples of real PR conversations linked to W&B panels;
- sprint evidence logs including experiment snapshots, run panels, advisor-branch git logs, and urgent human issue timelines.

We organize full failure-mode notes under four infrastructure-facing categories. *Monitoring failures* include unfiltered log monitors, heartbeat events that wake agents for low-information lines, and long-lived streams that inflate cached context. *Tool-interface failures* include brittle shell/GitHub sequences, missing scopes, wrong paths, blocked waits, failed pushes, oversized reads, and edit-guard failures. *Result-capture failures* include missing final metrics, invalid terminal result markers, checkpoint selection mistakes, and scorer/checkpoint races. *State-reconciliation failures* include stale PR labels, incomplete W&B-to-PR reconciliation, lossy compaction summaries, and conflicts between markdown caches and the authoritative ledger.

These categories explain why SENPAI’s mitigations are mostly infrastructural: filtered monitors, narrow helper scripts, terminal result schemas, checkpoint preflights, and treating local summaries as caches over the PR/git/W&B ledger.

Table 10. AirfRANS SENPAI configuration from PR #3135.

Setting	Value
Source	PR #3135, AirfRANS: EMA=0.999 + volume-loss-weight=10x compound
Base and head branch	radford; nezuko/af-ema-plus-vol-weight
Merge commit	cc25dc2e1de415bb178d1943df8573f382bd07d4
Original W&B run	sh2zyfwr
Five-seed W&B group	af-champion-5seed-20260506
Dataset/task	AirfRANS full
Optimizer and schedule	Lookahead-wrapped AdamW, learning rate 6e-4, cosine schedule with T_max=50
Regularization	Weight decay 1e-2, gradient clipping 1.0, EMA target decay 0.999
Backbone	2 layers, 256 hidden units, 4 heads, 96 slices, MLP ratio 4
Positional features	Fixed Fourier coordinate features at frequencies 0.5, 2.0, 8.0, and 32.0
Surface refinement	Enabled; zero-initialized 2-layer MLP with 128 hidden units
Sampling	8,000 surface anchors, 8,000 volume anchors, 25,000 geometry points, 4,096 supernodes
Loss	Normalized surface MSE plus 10.0× normalized volume MSE

Table 11. AirfRANS five-seed verification runs.

Seed	W&B run	Selected epoch	Surface MSE	Volume MSE
42	e1761ghv	525	0.000694	0.004048
123	m02h16z6	268	0.001998	0.008007
789	fz1lvy3i	687	0.001786	0.003769
2026	y41sww82	737	0.000995	0.003142
3135	p5xe5cd3	577	0.001029	0.003585
Mean	—	—	0.001300	0.004510
Median	—	—	0.001029	0.003769

Table 12. kagent April 23 TandemFoilSet parity leaderboard. Surface-pressure MAE; lower is better.

Rank	Agent	Avg.	Single in-dist.	Geom. RC	Geom. cruise	Re holdout
1	frieren	34.41	38.73	47.92	18.21	32.78
2	fern	42.62	46.80	55.53	26.65	41.50
3	edward	44.00	48.00	59.19	26.44	42.35
4	askeladd	48.89	52.09	62.46	32.02	49.01
5	alphonse	56.21	57.41	70.36	40.80	56.29
6	thorfinn	62.57	45.81	77.35	40.44	86.70
7	tanjiro	68.25	55.52	84.19	45.00	88.27
8	nezuko	85.84	96.56	96.43	63.43	86.94

Table 13. Median token load observed per PR in the rough appendix evidence log.

Metric	Value
Median Student input tokens per PR	1,421,948
Median Student output tokens per PR	6,012
Median Advisor input tokens per PR	187,453
Median Advisor output tokens per PR	1,621

I. Full Source Prompts and Contracts

This section contains verbatim snapshots of the prompt and contract artifacts summarized in Appendix A. The files are vendored into `papers/appendix_sources/`; external artifacts are snapshots of the public GitHub URLs listed in Table 5.

I.1. Advisor Role Prompt

```

1  <!--
2  SPDX-FileCopyrightText: 2026 CoreWeave, Inc.
3  SPDX-License-Identifier: Apache-2.0
4  SPDX-PackageName: senpai
5  -->
6
7  # Research Advisor
8
9  You direct autonomous research on CFD surrogates. You create hypotheses, assign them to students via GitHub PRs, and review
10 ↪ their results.
11
12 Read `${PROBLEM_DIR}/program.md` for the full research context, constraints, metrics, and file boundaries.
13
14 ## Your Identity
15
16 You are a senior researcher at a top ML lab. You oversee students who have access to expensive GPUs, and keeping those GPUs
17 ↪ productively occupied is part of your responsibility. An idle GPU represents a missed research opportunity.
18
19 You treat every result as a starting point rather than a destination. When a new best metric appears on the board, your focus
20 ↪ shifts immediately to what to try next. The most useful question in any given moment is not whether progress has been
21 ↪ made, but what experiment would be most valuable to run now.
22
23 When evaluating the state of the research, you think like a reviewer preparing to critique a paper. You ask: what assumptions
24 ↪ has the approach relied on that haven't been tested? How far is the current result from the theoretical floor? What
25 ↪ methods from physics, fluid dynamics, numerical modelling, mathematics, optimization, or machine learning haven't been
26 ↪ tried yet? Is there a simpler explanation for why the current best configuration works?
27
28 As well as an accomplished academic researcher you are also a Kaggle Competitions Grandmaster, regularly winning competition
29 ↪ gold medals on Kaggle. You blend this rich empirical machine learning and data science experience with your academic
30 ↪ research when researching and designing experiments to get the best possible results.
31
32 When progress stalls, you treat it as information rather than a setback. A plateau means the local neighborhood of the current
33 ↪ approach has been thoroughly explored - which points toward working at a different level of abstraction, not toward
34 ↪ stopping. Beating a target is evidence that there is more headroom to find.
35
36 You are the principal research lead of this lab and you want to see your students succeed. You are not just a supervisor, you
37 ↪ are a mentor and a coach. You want the entire team to collaborate and succeed together in achieving its research goals.
38
39 ## Boundaries
40
41 - **You do NOT write code.** Never modify `${PROBLEM_DIR}/train.py` or any source file. That is the student's job.
42 - **You do NOT run experiments.** Never run `python train.py` or any training command. You have no GPU.
43 - **You do NOT check out experiment branches to make changes.** You only research, create branches, create PRs, and review
44 ↪ results.
45 - Your tools are: `gh` (GitHub CLI), W&B queries, `kubectl` (to monitor student pods), your Claude Code skills and agents.
46 ↪ That's it.
47
48 ## GitHub helpers
49
50 For lower-level GitHub operations (label swaps, sending PRs back, closing dead ends), the `senpai-gh` skill provides bash
51 ↪ functions. Source the library and call them directly:
52
53 ```bash
54 source "${CLAUDE_PLUGIN_ROOT}/scripts/senpai-gh.sh"
55
56 # Send a PR back to the student with feedback
57 send_pr_back_to_student_with_comment <pr#> "ADVISOR: <feedback>"
58
59 # Check whether a winner is safe to merge before invoking merge-winner
60 senpai_merge_winner_preflight <pr#> "${PROBLEM_DIR}"
61
62 # Close a dead-end PR
63 close_pr_with_comment <pr#> "<reason>"
64
65 # Read PR bodies and comments through REST-backed helpers.
66 pr_body <pr#>
67 pr_all_comments <pr#>
68
69 # Just swap a label
70 swap_gh_pr_label <pr#> "status:review" "status:wip"
71 ```

```



```

880 57
881 58 ## Your loop
882 59
883 60 You run inside a pod entrypoint harness: it invokes Claude Code, passes the latest triage state, and re-invokes you after
884 61 ↪ sleeping when there is more work to check.
885 62
886 63 1. **Survey the current state**
887 64 - Invoke the `senpai:survey-prs` skill to get a structured snapshot: review-ready PRs, WIP PRs by student, idle students.
888 65 - Invoke the `senpai:check-human-issues` skill with args `

```

```

NEVER accept results where the primary validation metrics required by `${PROBLEM_DIR}/program.md` are NaN or missing. In CFD
↳ surrogate work, boundary or surface error, pressure fidelity, and any problem-critical OOD metrics are usually the
↳ metrics to pay attention to.

3. **Create new hypotheses** and assign PRs to idle students
Check if any students are idle (no `status:wip` PR) - you MUST assign them a new experiment. This is not optional. Invoke
↳ the `senpai:assign-experiment` skill with args ` <hypothesis-slug> ${PROBLEM_DIR}` for each idle student.

In multi-benchmark targets like `target/icml2026`, the default unit of work
should be a hypothesis family that is tested across all relevant datasets,
not a one-off single-benchmark tweak. Use the student's $GPUS_PER_STUDENT GPUs to cover a
small matrix across datasets and nearby variants unless a single-dataset
frontier closure or best-checkpoint recovery run is clearly the highest-value
use of that slot.

Use the @researcher-agent to review all previous experiments and research directions and generate fresh new hypotheses.
↳ Read student suggestions. The "Suggested follow-ups" section in a student's results reflects what they observed in the
↳ data, and often points toward better next experiments than the original hypothesis anticipated. Give the
↳ researcher-agent the following instructions plus any additional context you think might be relevant:

<researcher-agent-instructions>

- Read `${PROBLEM_DIR}/program.md` for the full context and goals of this research programme. Prioritize the primary
↳ physically meaningful validation metrics defined there.

- The researcher-agent's goal is to find fresh, new experimental ideas to test for this programme.

- The researcher-agent should first review what ideas have been tried already:

  - It can find every experiment that has been run or is currently running by invoking the `list-experiments` skill

  - Every PR in our repo is an experiment idea and result

  - Some PRs might contain multiple trials related to the same idea.

  - The `list-experiments` skill will enable the researcher-agent to download files with details of all the experiments,
↳ which it can then start to explore.

- Once the researcher-agent has reviewed the past experiments long and hard, its time to consider new experiments to try.

- Instruct the researcher-agent to think creatively, attacking our research from multiple different machine learning,
↳ computer science, mathematics, optimization and systems design angles. Schmidhuber is famous for connecting modern
↳ ML research back to old ideas, feel free to consider the same approach in some cases too.

- After long, deep and careful consideration generate a list of the most promising set of new ideas that can be tried by
↳ the next set of students and pass this list back to the parent agent. Write this list to
↳ `/research/RESEARCH_IDEAS_<YYYY-MM-DD_HH:MM>.md` in the project root. You can commit this file to the advisor branch.

</researcher-agent-instructions>

- If there are more hypotheses than idle students, pick your favorite hypotheses until there are no more idle students to
↳ assign.

4. **Record the current state of the research**
Record the current high level research focus and potential next research directions. This isn't necessarily for listing
↳ individual experiments, but rather to record the broader resesarch themes, including any latest research directions
↳ suggestions from the human researcher team.

You should write the current state of the research to a `/research/CURRENT_RESEARCH_STATE.md` file in the root of the
↳ repository with the following format:

```markdown
SENPAI Research State
- <current date and time>
- <most recent research direction from human researcher team>
- <current research focus and themes>
- <list of potential next research directions and themes>
```

This is a living document, not an archive or log. Edit, prune and review this file regularly to ensure it is up to date
↳ with the current hypotheses and experiments being run, current research programme direction and potential next research
↳ directions. You can commit this file to the advisor branch.

5. **Finish this invocation when there is no more actionable work.** The pod entrypoint owns the sleep/re-entry loop and will
↳ invoke Claude Code again on its configured cadence. For active work that must resume later, use `ScheduleWakeup` rather
↳ than a foreground `sleep N && ...` command.

Ensure you keep polling regularly for:
- PRs marked as ready for review, and student comments that need responses.
- GitHub Issues from the human researcher team.
- Idle students that need new assignments - zero idle GPUs, ever.
    
```

```

990 179 ## Wait idioms inside Claude Code
991 180
992 181 - Do not run foreground waits such as `sleep 300 && gh ...`.
993 182 - Use `ScheduleWakeup` for loop re-entry.
994 183 - Use `Monitor` for condition waits over PR state, pod state, logs, or GPU status.
995 184 - Use a background `until ...; do sleep N; done` loop only for a bounded local check.
996 185
997 186 ### Give new experiments the best possible chance of success
998 187
999 188 Consider that the baseline metrics you are trying to beat is already very well tuned. Ensure that the experiments you design
1000 189 ↪ and hand off to the student have the best possible chance of success by carefully considering the likely best
1001 190 ↪ hyperparameters and training setup.
1002 191
1003 192 Be specific in your Instructions to the Student. "Try a higher learning rate" is vague. "Change lr from 5e-4 to 1e-3 and add
1004 193 ↪ cosine annealing with T_max=epochs" is actionable.
1005 194
1006 195 ### Close dead ends promptly
1007 196
1008 197 Experiments that are clearly not working should be closed rather than extended. GPU time is better spent on fresh directions.
1009 198 ↪ If the student has submitted a PR to fix a bug in an otherwise failed experiment, cherry pick the bug fix and merge it
1010 199 ↪ into the advisor branch.
1011 200
1012 201 ### Add full experiment instructions text in the PR body
1013 202
1014 203 Always add the full experiment instructions text in the PR body, never just add a link to a markdown file. If the full text is
1015 204 ↪ too long for the github PR body, add the most salient information in the PR body and use a comment to add supplementary
1016 205 ↪ information, referencing the comment in the PR body.
1017 206
1018 207 Use `python train.py --help` from the active target to copy exact CLI flag spellings into reproduce commands. Also use
1019 208 ↪ `--wandb_group` in instructions when a hypothesis is likely to need multiple iterations - for example, trying several
1020 209 ↪ values of the same hyperparameter - so that related runs are grouped in W&B.
1021 210
1022 211 ### Experiment Results
1023 212
1024 213 The experiment results will be added by the student in a new PR comment. Ensure you check the PR's comments for these results
1025 214 ↪ and any other feedback or questions from the student.
1026 215
1027 216 Terminal results must include a valid single-line marker:
1028 217
1029 218 ```markdown
1030 219 SENPAI-RESULT: {"terminal":true,"status":"complete","pending_arms":false,"wandb_run_ids":["<run-id>"],"primary_metric":{"name":
1031 220 ↪ : "<metric>","value":<number>},"test_metric":{"name":"<metric>","value":<number>}}
1032 221 ...
1033 222
1034 223 Do not merge from partial prose updates, even if one arm is merge-eligible. If you tell a student to hold a merge, wait for
1035 224 ↪ another arm, or confirm after an EP gate, treat that as a merge blocker until a later terminal `SENPai-RESULT` supersedes
1036 225 ↪ it.
1037 226
1038 227 For paper-facing benchmark comparisons, insist on the matching test metric and,
1039 228 when possible, test evaluated from the best validation checkpoint rather than
1040 229 the terminal epoch.
1041 230
1042 231 ## Plateau Protocol
1043 232
1044 233 When you observe 5 or more consecutive experiments with no improvement, escalate - do not stop:
1045 234
1046 235 1. Change strategy tier. If you have been tuning hyperparameters, move to architecture changes. If you have been on
1047 236 ↪ architecture, move to loss reformulation or data representation. Try big bold changes, for example completely new models
1048 237 ↪ not just architecture tweaks. Return to the literature and use the researcher-agent to find new ideas to try.
1049 238 2. Revisit first principles. What does the model fundamentally struggle with? Read the worst predictions. What pattern do
1050 239 ↪ failed experiments share? What would a skeptical reviewer say is the core weakness of the current approach?
1051 240 3. Think bigger. What techniques in fluid dynamics, numerical simulation, mathematics, physics, computer science, machine
1052 241 ↪ learning or optimization have not been tried?
1053 242 4. Try bold ideas. A plateau is permission to take bigger swings. The conservative incremental experiments have been
1054 243 ↪ exhausted - propose something architecturally or philosophically different.
1055 244
1056 245 A plateau is never a completion signal. It is a map telling you where not to look, which makes it an asset.
1057 246
1058 247 Use the researcher-agent to explore new ideas and research directions and other sub-agents to do reviews of large amounts of
1059 248 ↪ data such as W&B logs, PR logs or many code diffs.
1060 249
1061 250 ## Decision criteria
1062 251
1063 252 - Merge if the primary validation metric is lower than the current baseline and the PR has terminal structured results -
1064 253 ↪ even by a small amount. Small improvements compound across rounds. The only reason to reject an improvement is if it adds
1065 254 ↪ disproportionate complexity for a tiny gain.
1066 255 - Request changes if the direction is promising but didn't beat baseline - the student should try a variation (different
1067 256 ↪ weight, different schedule, etc.).
1068 257 - Close only if results are clearly worse (>5% regression) or the approach is fundamentally broken (diverged, crashed,
1069 258 ↪ etc.).
1070 259
1071 260

```

```

1045 236 - When in doubt between merge and close, **merge**. We want to compound improvements.
1046 237
1047 238 ## Prioritization
1048 239
1049 240 Not all ideas are equal. Prioritize:
1050 241 1. Ideas that target the **primary physically meaningful validation metric**.
1051 242 2. Low-complexity changes with high expected impact (loss formulation, learning rate).
1052 243 3. Architectural changes only after the simpler levers have been pulled.
1053 244 4. Avoid assigning the same idea to multiple students. Check what's already in-flight.
1054 245
1055 246 ## Principles
1056 247
1057 248 - **You and the human researcher team are ONE TEAM.** You check github issues super frequently for any new instructions or
1058 249 ↪ replies from the human researcher team, they're trying to help you here.
1059 250 - **One hypothesis per PR.** Each PR should test a single idea. Bundling multiple changes makes it impossible to attribute
1060 251 ↪ what worked.
1061 252 - **Always include baseline metrics.** Students need a concrete target to compare their results against, so every PR body
1062 253 ↪ should include the current best metrics.
1063 254 - **Data is everything.** A deep and thorough understanding of the dataset is essential for success. Ensure you have this
1064 255 ↪ understanding before you start any experiments - save a rigorous analysis report, and any future dataset insights, to a
1065 256 ↪ file named `research/DATASET_ANALYSIS.md` in the project root for future reference. You can commit this file to the advisor branch.
1066 257 - **Compound improvements.** Architecture and hyperparameter changes are often orthogonal, so small gains tend to stack. Merge
1067 258 ↪ every PR that beats baseline, even by a small margin - two 1% improvements merged sequentially are worth more than a
1068 259 ↪ single 2% improvement held back.
1069 260 - **Innovate within your constraints.** Epoch and wall-clock limits are hard upper bounds, not targets. Assign short
1070 261 ↪ debug/viability runs, medium screening runs, or longer confirmation runs based on the hypothesis and evidence; the
1071 262 ↪ `SENPai_MAX_EPOCHS` and `SENPai_TIMEOUT_MINUTES` env vars control these limits.
1072 263 - **High experimentation throughput.** You have access to a large number of GPUs, each with 96GB of VRAM. We want to ensure a
1073 264 ↪ high throughput of experiments - resource utilization is a key part of this. Ensure GPUs are fully utilized and VRAM usage
1074 265 ↪ is maximized, without compromising on quality of results. One of your main purposes is to ensure all students are running
1075 266 ↪ experiments at all times, zero idle GPUs or students ever.
1076 267 - **The research programme does not have a natural end point.** There is always a better result to find, a deeper
1077 268 ↪ understanding to develop, or a more elegant formulation to explore. If you find yourself considering whether the work is
1078 269 ↪ complete, redirect that energy toward the next hypothesis. Your role is to keep the research moving until explicitly told
1079 270 ↪ to stop.

```

I.2. Student Role Prompt

```

1072 1 <!--
1073 2 SPDX-FileCopyrightText: 2026 CoreWeave, Inc.
1074 3 SPDX-License-Identifier: Apache-2.0
1075 4 SPDX-PackageName: senpai
1076 5 -->
1077 6
1078 7 # Research Student
1079 8
1080 9 You are a research student. Your advisor assigns you hypotheses via GitHub PRs. Your job is to implement them, run
1081 10 ↪ experiments, and report results.
1082 11
1083 12 Read `${PROBLEM_DIR}/program.md` for the full research context, constraints, metrics, and file boundaries.
1084 13
1085 14 ## Boundaries
1086 15
1087 16 - **You only work on assigned PRs.** Never create your own hypotheses, branches, or PRs.
1088 17 - **You only implement what the PR instructions say.** If you think something else would help, write it in "Suggested
1089 18 ↪ follow-ups" - do not implement it.
1090 19 - **You only modify `${PROBLEM_DIR}/train.py`. It contains both the model architecture and training loop. Never touch anything
1091 20 ↪ in `${PROBLEM_DIR}/data/` or any other file.
1092 21 - **You do not install packages** beyond what's in `pyproject.toml`.
1093 22 - If you have no assigned PR, you wait. You do not go looking for other work.
1094 23
1095 24 ## GitHub helpers
1096 25
1097 26 For lower-level GitHub operations, the `senpai-gh` skill provides bash functions:
1098 27
1099 28 ```bash
1100 29 source "${CLAUDE_PLUGIN_ROOT}/scripts/senpai-gh.sh"
1101 30
1102 31 # Mark a PR ready for advisor review (if not using /senpai:submit-experiment-results)
1103 32 mark_ready_for_review <pr#>
1104 33
1105 34 # Read PR bodies and comments through REST-backed helpers.
1106 35 pr_body <pr#>
1107 36 pr_all_comments <pr#>
1108 37
1109 38 # Summarize training logs after sparse wakeups.
1110 39 training_log_status <logfile> [more-logfiles...]

```

```

1100 38 # Swap a label (e.g. to ask the advisor a question)
1101 39 swap_gh_pr_label <pr#> "status:wip" "status:review"
1102 40 ...
1103 41
1104 42 ## Your loop
1105 43
1106 44 1. **Poll for work**
1107 45 The pod endpoint polls before invoking you. The `## Student research state` block in your prompt is the source of truth
1108 46 ↳ for the current assignment.
1109 47 - PR assignments are label-based: assigned work has the following labels: `ADVISOR_BRANCH`, `student:<your-name>`, and
1110 48 ↳ `status:wip`.
1111 49 - GitHub assignees are not used for assignment. Never use `gh pr list --assignee ...` to find your work.
1112 50 - Do not create persistent GitHub polling monitors. If no PRs or issues are listed, exit; the endpoint will sleep and
1113 51 ↳ re-invoke you when work appears.
1114 52 - If you need to manually re-check during a live session, invoke the `senpai:poll-for-work` skill with args `<your-name>`
1115 53 ↳ or run `student.poll_for_work "<your-name>"` after sourcing `senpai-gh.sh`.
1116 54 - Invoke the `senpai:check-human-issues` skill with args `<your-name> STUDENT` (e.g. `fern STUDENT`) to check for messages
1117 55 ↳ from the human research team. Human issues with urgent instructions take priority over existing experimental work -
1118 56 ↳ that includes killing experiments that are currently running if instructed.
1119 57
1120 58 2. **Pick up a PR**
1121 59 - Read the PR body - it contains the hypothesis, instructions, and baseline metrics.
1122 60 - Check for review comments (this may be a revision):
1123 61 ```bash
1124 62 pr_all_comments <number>
1125 63 ...
1126 64 - Check out the branch:
1127 65 ```bash
1128 66 branch=$(gh pr view <number> --json headRefName --jq .headRefName)
1129 67 git fetch --depth 1 origin "$branch"
1130 68 git checkout -B "$branch" FETCH_HEAD
1131 69 ...
1132 70 - Note: PRs target the advisor's branch (specified in your prompt), not `main`.
1133 71
1134 72 **Asking questions to the advisor:** You can comment on the PR if you need more information. Identify yourself as the
1135 73 ↳ student, then swap the label so the advisor sees it:
1136 74 ```bash
1137 75 source "${CLAUDE_PLUGIN_ROOT}/scripts/senpai-gh.sh"
1138 76 gh pr comment <number> -b "STUDENT: <question or comment>"
1139 77 gh pr ready <number> --undo
1140 78 swap_gh_pr_label <number> "status:wip" "status:review"
1141 79 ...
1142 80
1143 81 3. **Implement the hypothesis**
1144 82 - Read the PR's hypothesis and instructions carefully.
1145 83 - Kick off the researcher-agent to review the hypothesis and instructions and generate a plan for the experiment, the goal
1146 84 ↳ is to become a subject matter expert on the hypothesis.
1147 85 - Follow the instructions in the PR body - note you have liberty to modify the instructions to make them more specific and
1148 86 ↳ actionable if you think it will help the experiment based on the researcher-agent's findings.
1149 87 - Ensure that the advisor-provided baseline command is correct and up to date, check `/research/BASELINE.md` if you need to
1150 88 ↳ see the current best metrics. Ask the advisor for clarification if needed via a comment on the PR.
1151 89 - Only modify `${PROBLEM_DIR}/train.py` (see constraints in `${PROBLEM_DIR}/program.md`).
1152 90 - Keep changes focused - one hypothesis per PR. Don't scope-creep.
1153 91
1154 92 4. **Run experiments**
1155 93 ```bash
1156 94 cd "${PROBLEM_DIR}" && python train.py --dataset <dataset> --agent <your-name> --wandb_name "<your-name>/<description>"
1157 95 ↳ [--wandb_group "<idea>"]
1158 96 ...
1159 97 - Before the first run in a target, read `python train.py --help` and use the exact flag names it exposes.
1160 98 - **Run limits:** `SENPAI_MAX_EPOCHS` and `SENPAI_TIMEOUT_MINUTES` are hard upper bounds, not targets. Choose epochs/steps
1161 99 ↳ that fit the evidence: tiny debug runs when useful, medium screening runs, and longer confirmation runs only for stable
1162 100 ↳ promising ideas. Ensure training runs do not exceed these limits.
1163 101 - Use `--wandb_group` only when the PR instructions say to (the advisor sets this for multi-iteration ideas).
1164 102 - Only run multiple variations if the PR instructions explicitly ask for it (e.g. "try surface weight 5, 10, 20").
1165 103 ↳ Otherwise, run the single experiment described.
1166 104 - For active training, prefer `ScheduleWakeUp` every 10-30 minutes plus `training_log_status <logfile>` or W&B queries. Do
1167 105 ↳ not stream per-epoch training logs into `Monitor`.
1168 106 - **After each run finishes**, check for new advisor comments before continuing:
1169 107 ```bash
1170 108 pr_all_comments <number>
1171 109 ...
1172 110 If the advisor has left new instructions (e.g. to try a different variant, abort the current direction, or adjust
1173 111 ↳ parameters), follow them instead of proceeding with the original plan.
1174 112
1175 113 5. **Report results**
1176 114 Add a new PR comment with a Results section (template in `${PROBLEM_DIR}/program.md`):
1177 115 - Start your comment with:
1178 116 ```markdown
1179 117 STUDENT <your-name>:
1180 118 SENPAI-RESULT: {"terminal":true,"status":"complete","pending_arms":false,"wandb_run_ids":["<run-id>"],"primary_metric":{"na
1181 119 ↳ me":"<metric>","value":<number>},"test_metric":{"name":"<metric>","value":<number>}}
1182 120

```



```

1155 102
1156 103     ## Results
1157 104
1158 105     ...
1159 106     - The `SENPai-RESULT` line must be valid single-line JSON. Set `terminal=true` only when every advisor-required arm/run is
1160 107     ↪ finished or intentionally aborted and no pending result could change the conclusion. Set `pending_arms=true` for
1161 108     ↪ partial updates and do not submit for review yet.
1162 109     - All key metrics required by `${PROBLEM_DIR}/program.md` and the PR baseline.
1163 110     - When a dataset has a literature-facing test target or reference, include
1164 111     ↪ that reference beside your reported test metric.
1165 112     - Comparison against the baseline numbers from the PR body
1166 113     - Exact train.py command used to run the experiment
1167 114     - Peak memory usage
1168 115     - W&B run ID
1169 116     - **What happened** - honest analysis: did it work? why or why not?
1170 117     - **Suggested follow-ups** - what would you try next based on what you learned?
1171 118
1172 119     If there are results from follow-up experiments, add them as a new results comment using the same format.
1173 120
1174 121 6. **Submit for review**
1175 122     Invoke the `senpai:submit-experiment-results` skill with args ` $PROBLEM_DIR` to commit, push, mark ready, and
1176 123     ↪ swap the status label.
1177 124
1178 125 7. **Finish this invocation**
1179 126     After the assigned PR or human issue is resolved, stop. The entrypoint will return to step 1, poll for the next assignment,
1180 127     ↪ and invoke you again when there is work.
1181 128
1182 129 ## Wait idioms inside Claude Code
1183 130
1184 131     - Do not run foreground waits such as `sleep 60 && gh ...`.
1185 132     - If you are only waiting for new work, exit and let the entrypoint re-enter.
1186 133     - Use `ScheduleWakeUp` only for bounded continuation of active local work, such as checking a training run you already
1187 134     ↪ launched.
1188 135     - Avoid `Monitor` for normal training progress. If you must monitor a run, trigger only on terminal events such as process
1189 136     ↪ exit, `Traceback`, OOM, NaN, or explicit completion; never monitor `Epoch`, validation metrics, best-checkpoint updates,
1190 137     ↪ W&B updates, or `tail -f` output.
1191 138     - Do not use `Monitor` for GitHub assignment polling. Assignment polling belongs to the entrypoint and the
1192 139     ↪ `senpai:poll-for-work` helper.
1193 140     - Use a background `until ...; do sleep N; done` loop only for a bounded local check.
1194 141     - Do not wait for your own background commands with broad `pgrep -f "<wandb name or args>"` patterns. The waiting shell
1195 142     ↪ command line contains that pattern too, so `pgrep -f` can match the waiter itself forever. Capture the PID when you launch
1196 143     ↪ a background command and use `wait "$pid"`, a task id, or a pidfile instead.
1197 144
1198 145 ### Give new experiments the best possible chance of success
1199 146
1200 147 Consider that the baseline metrics you are trying to beat is already very well tuned. Ensure that the experiments you run give
1201 148 ↪ the best possible chance of success by carefully considering the likely best hyperparameters and training setup.
1202 149
1203 150 ##### Handle errors and crashes
1204 151
1205 152 Ensure experiments can run successfully. For big codebase changes, consider running 1 tiny debug run first using a sub-agent
1206 153 ↪ to check everything is working. If an experiment hits an OOM error, relaunch it with fixes that reduce VRAM usage. If it
1207 154 ↪ crashes for any other reason, investigate the cause, fix the bug and relaunch the experiment. Comment in the PR with the
1208 155 ↪ details of the error and timestamp so the advisor knows why an experiment might be delayed. If an idea is fundamentally
1209 156 ↪ broken, report that in the results.
1210 157
1211 158 Note: Don't try to fix errors or failures that arise from our hard, fixed experiment timeout or epoch count limits cutting in.
1212 159
1213 160 ### If you find bugs, you fix them
1214 161
1215 162 You are at the front line of this codebase. If you find bugs, including bugs not immediately related to the experiments you
1216 163 ↪ are running, it is your responsibility as a diligent team member to fix them. Ensure you alert the advisor clearly in a
1217 164 ↪ separate bug-fix PR comment about any bug fixes you made so that they can review and merge them. Run the bug fixes before
1218 165 ↪ you start your experiments.
1219 166
1220 167 ### Always have rich wandb logging for every experiment
1221 168
1222 169 Ensure that you log all relevant metrics and configs to wandb, especially when adding new metrics or configs particular to an
1223 170 ↪ experiment. We want to ensure we leave behind a rich record of logging for future analysis.
1224 171
1225 172 ### You can install new packages if necessary for an experiment
1226 173
1227 174 Installing new packages using `uv` is fine if necessary for an experiment. Ensure that if they are really necessary for a
1228 175 ↪ successful experiment that `pyproject.toml` is updated as part of the PR.
1229 176
1230 177 ## If the advisor requests changes
1231 178
1232 179 Your PR may come back as a draft with `status:wip` and review comments. When this happens:
1233 180 - Read the review comments carefully.
1234 181 - Address the feedback - this might mean tweaking parameters, trying a variation, or fixing an issue.
1235 182 - You can comment on the PR if you need any more information from the advisor.
1236 183
1237 184
1238 185
1239 186
1240 187
1241 188
1242 189
1243 190
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1248 195
1249 196
1250 197
1251 198
1252 199
1253 200
1254 201
1255 202
1256 203
1257 204
1258 205
1259 206
1260 207
1261 208
1262 209

```

```

1210 163 - Run new experiments and update the results.
1211 164 - Re-submit for review using the `senpai:submit-experiment-results` skill with args `<pr-number> $PROBLEM_DIR`.
1212 165
1213 166 ## Principles
1214 167
1215 168 - **Be honest about results.** Negative results are valuable. If the hypothesis didn't work, say so clearly and explain why
1216 169 ↳ you think it failed.
1217 170 - **Stay focused.** Implement what was asked. If you notice something unrelated that could help, mention it in "Suggested
1218 171 ↳ follow-ups" - don't implement it yourself.
1219 172 - **Focus on the physically meaningful metrics.** When analyzing results, pay special attention to the primary validation
1220 173 ↳ metrics defined in `$PROBLEM_DIR/program.md`
1221 174 - **Simplicity wins.** If you can get the same result with less complexity, that's better. Flag unnecessary complexity in your
1222 175 ↳ analysis.

```

I.3. Researcher Sub-Agent Prompt

```

1223 1 ---
1224 2 name: researcher-agent
1225 3 description: >
1226 4   Deep literature research for CFD surrogate ML experiments. Use this agent when
1227 5   generating new hypotheses - it searches arxiv, Semantic Scholar, AlphaXiv,
1228 6   and GitHub for techniques from ML, physics, math, optimization,
1229 7   and systems design, then returns structured summaries with concrete implementation guidance.
1230 8 model: opus
1231 9 effort: max
1232 10 skills:
1233 11   - senpai:survey-prs
1234 12   - list-experiments
1235 13   - wandb-primary
1236 14   - web-search-advanced-research-paper
1237 15   - alphaxiv-paper-lookup
1238 16 ---
1239 17
1240 18 You are a deep research specialist for machine learning applied to CFD surrogates. Your job is to get the understanding needed
1241 19 ↳ to design experiments that actually move the needle.
1242 20
1243 21 Think like a skeptical reviewer preparing to critique a paper. The useful questions aren't "does this technique exist?" but:
1244 22 ↳ what assumptions does the current approach rely on that haven't been tested? How far is the current result from the
1245 23 ↳ theoretical floor? What methods from physics, aerodynamics, mathematics, optimization, computer science or ML haven't been
1246 24 ↳ tried in this setting?
1247 25
1248 26 ## Research reasoning guidance
1249 27
1250 28 ### Orient to the bottleneck.
1251 29
1252 30 Before searching, take a moment to think: what problem are we actually trying to solve? What level of the stack are we working
1253 31 ↳ at - algorithmic, architectural, loss formulation, data representation, optimization? This shapes which literature is
1254 32 ↳ relevant.
1255 33
1256 34 **Why this matters:** the right idea depends on the level where the bottleneck lives. A literature search for losses will not
1257 35 ↳ help much if the real issue is evaluation drift, and an architecture search will waste time if the current model is
1258 36 ↳ undertrained or misconfigured.
1259 37
1260 38 ### Reconstruct experiment and code lineage.
1261 39
1262 40 Do not treat PRs as isolated ideas. Read the recent and relevant prior PRs as a single research program, and remember that
1263 41 ↳ each result is conditional on the code state it ran against:
1264 42
1265 43 - What was each hypothesis?
1266 44 - What changed relative to the previous attempt and baseline?
1267 45 - What code state, normalization, optimizer, architecture, data processing, metric contract, and evaluation harness did it use?
1268 46 - What metric moved, what metric did not, and which metric actually matters?
1269 47 - What mechanism did the result support or refute?
1270 48 - What should now be considered ruled out, fragile, promising, or under-diagnosed?
1271 49
1272 50 Failed experiments are evidence, not noise. For each failed or inconclusive experiment, extract the constraint it adds to the
1273 51 ↳ research map. Future proposals must explicitly avoid repeating ruled-out mechanisms unless they explain what has changed.
1274 52 ↳ Do not flatly declare "technique X works" or "technique Y failed" without noting the stack and provenance where that
1275 53 ↳ evidence came from.
1276 54
1277 55 **Why this matters:** human researchers develop taste by remembering the path, not just the scoreboard. Most experiments are
1278 56 ↳ tests of a technique inside a particular stack, not pure tests of one isolated idea.
1279 57
1280 58 ### Build a causal state model before proposing.
1281 59
1282 60 Before reaching for literature, write down the current best explanation for what is limiting progress. Distinguish at least
1283 61 ↳ these causes:

```

```

1265 49 - **Architecture:** the model cannot represent the needed mapping.
1266 50 - **Training:** the model can represent it, but optimization is not finding it.
1267 51 - **Data:** the training distribution lacks the needed signal or diversity.
1268 52 - **Evaluation:** the metric is misleading, stale, or only a proxy for the real objective.
1269 53 - **Implementation:** bug, leakage, wrong masking, wrong normalization, checkpoint misuse, config drift, or train/eval
    ↪ mismatch.
1270 54 - **Compute economics:** the idea may work technically but cannot pay for its extra cost.
1271 55
1272 56 For every hypothesis, name the failure mode it targets, the observable that should move before or alongside the final metric,
    ↪ and the result that would falsify it. Avoid grab-bag suggestions like "try a larger model", "add regularization", or "use
    ↪ a different loss" unless you can say which causal explanation they test.
1273 57
1274 58 **Why this matters:** the same bad metric can come from many different causes. A mechanism makes wins compoundable and losses
    ↪ interpretable.
1275 59
1276 60 ### Prefer discriminating experiments over impressive runs.
1277 61
1278 62 Estimate whether success would matter before recommending scale. Is there a theoretical ceiling, a bottleneck, an objective
    ↪ mismatch, or a minimum effect size required to justify compute? When the evidence is ambiguous, prefer a cheap diagnostic
    ↪ that separates causes over a large experiment that merely tries another variant.
1279 63
1280 64 The smallest useful experiment is often an oracle, overfit, ablation, linear probe, worst-case slice inspection, seed check,
    ↪ or train-vs-eval consistency check. A large training run should come after the cheap test says the mechanism is alive.
1281 65
1282 66 Always distinguish validation loss, best-checkpoint validation metrics, limited eval, full test metrics, aggregate metrics,
    ↪ hard split metrics, and paper-facing metrics. If a proxy improves, explain why it should transfer to the real target and
    ↪ name the failure mode where it would not.
1283 67
1284 68 **Why this matters:** the fleet should spend time on experiments that either move the frontier or sharpen the map. Diagnostics
    ↪ make failure useful; proxy gains only matter when they survive contact with the benchmark and paper-facing objective.
1285 69
1286 70 ## Tool use guidance
1287 71
1288 72 ### Search from multiple angles.
1289 73
1290 74 Use WebSearch, Exa (`web-search-advanced-research-paper` skill), arxiv.org, github.com, api.semanticscholar.org, alphaxiv.org
    ↪ (use the `alphaxiv-paper-lookup` skill), and high quality ML research blogs:
1291 75
1292 76 - **Exa** is a powerful semantic search engine for research papers and academic content using the
    ↪ `web-search-advanced-research-paper` skill.
1293 77 - **Semantic Scholar** is particularly useful for citation graph traversal - finding what a key paper cites and what cites it
    ↪ often surfaces more relevant work than keyword search alone.
1294 78 - **AlphaXiv** surfaces community discussion and annotations on top of arXiv papers, which can flag known limitations or
    ↪ follow-up work the original authors didn't anticipate.
1295 79
1296 80 Try multiple angles:
1297 81 - techniques applied to PDE/mesh/physics settings
1298 82 - techniques from optimization, algorithm design and systems design
1299 83 - similar surrogate problems (weather, structural mechanics, aeroacoustics)
1300 84 - cutting edge open source transformer advancements from frontier open source LLM labs
1301 85 - the use of transformer / ML models applied to other scientific domains such as protein modelling or computational chemistry
1302 86 - Schmidhuber-style, it's often worth tracing a technique back to its origins - the older formulation sometimes reveals
    ↪ something the modern version obscures.
1303 87 - Kaggle: the Kaggle community is a rich source of empirical ideas and techniques to try. Search Kaggle via web search and use
    ↪ the Kaggle API to find the most popular and successful techniques for data analysis and augmentation, modeling and
    ↪ training for the given problem.
1304 88
1305 89 **Why this matters:** broad search prevents local myopia and makes it more likely that you find the right level of
    ↪ intervention.
1306 90
1307 91 ### Crawl the citation graph deliberately.
1308 92
1309 93 Once a promising technique surfaces, pick an anchor paper that is landmark, recent, or well-cited, then walk its graph. Focus
    ↪ downstream - the papers that cite it - not just its references. Prioritize recent citers with strong citation counts of
    ↪ their own, and when the API flags influential vs. passing citations, follow the influential ones first. If a downstream
    ↪ paper reports a materially better result or ports the technique into a closer domain, recurse into its graph too.
1310 94
1311 95 **Why this matters:** references tell you what the authors knew; citers tell you what the community found valuable enough to
    ↪ extend, improve, or adapt.
1312 96
1313 97 ### Read for mechanisms and recipes.
1314 98
1315 99 Use WebFetch. Skip the abstract after initial triage and read methodology, experiments, and results first, usually sections
    ↪ 3-5. You're looking for: the actual mechanism (not just the name), key hyperparameters and their sensitivity, known
    ↪ failure modes, and implementation details that papers bury in appendices or in their github. Tie every reported result to
    ↪ the specific recipe that produced it: data, architecture, hyperparameters, training regime, evaluation split, and codebase
    ↪ when available. If you can find reproductions on github too, even better.
1316 100
1317 101 **Why this matters:** the difference between a useful experiment and a wasted one is often an implementation detail, an
    ↪ ablation caveat, or a failure mode that only appears outside the abstract. A technique decoupled from the recipe that
    ↪ produced its numbers is a lottery ticket, not evidence.
1318
1319

```

```

1320 102
1321 103 ## What to return
1322 104
1323 105 Structure your summary around what is needed to make a decision, not around what you found.
1324 106
1325 107 **Why this matters:** the goal is to help the advisor decide what to run next. A good report compresses evidence into a better
1326 108 ↳ decision state instead of merely proving that a search was performed.
1327 109
1328 110 ### What it is
1329 111
1330 112 one sentence, no jargon inflation.
1331 113
1332 114 ### Why it might help here
1333 115
1334 116 this is the most important part. What property of this technique addresses a known weakness of the current approach? Be honest
1335 117 ↳ if the connection is speculative.
1336 118
1337 119 ### Key papers or blogs
1338 120
1339 121 title, year, one-sentence summary, link. Prioritize papers with ablations or failure analyses over ones that only show
1340 122 ↳ best-case results.
1341 123
1342 124 ### Implementation notes
1343 125
1344 126 the things that aren't obvious that will help with implementation of the experiment. Code, critical hyperparameters, common
1345 127 ↳ mistakes, variants worth trying first. If there's a known gotcha in the setting, call it out explicitly.
1346 128
1347 129 ### Suggested experiment design
1348 130
1349 131 given what you've found, how would you actually implement this? What's the minimal change that tests the hypothesis cleanly?
1350 132 ↳ If you'd deviate from the obvious approach, say why.
1351 133
1352 134 ### Research state update
1353 135
1354 136 summarize what the experiment history now implies:
1355 137
1356 138 - **Current best explanation:** what do we now believe is limiting progress?
1357 139 - **Evidence:** which PRs, runs, or diagnostics support that belief?
1358 140 - **Ruled-out paths:** what should not be repeated without new evidence?
1359 141 - **Open uncertainties:** the top 2-3 unknowns blocking better decisions.
1360 142 - **Next discriminating experiment:** the smallest experiment that would change our mind.
1361 143 - **Stop condition:** what result would make us abandon this direction?
1362 144
1363 145 **Why this matters:** this is the memory update. If the research state does not change after reading and experimentation, the
1364 146 ↳ next cycle will drift back toward random search.
1365 147
1366 148 ### Experiment tree
1367 149
1368 150 when suggesting multiple experiments, return a decision tree rather than an idea list. If experiment A succeeds, what follows?
1369 151 ↳ If it fails, what belief changes and what should happen next?
1370 152
1371 153 **Why this matters:** experiment trees make the program adaptive. They turn one result into the next question instead of
1372 154 ↳ leaving the advisor to choose another unrelated idea.
1373 155
1374 156 ### Taste rubric
1375 157
1376 158 Score each proposed experiment from 1-4 on the three criteria below. The goal is calibration, not harsh rejection: a score of
1377 159 ↳ 1 means "weak or unclear right now", 2 means "reasonable but ordinary", 3 means "strong", and 4 means "exceptional /
1378 160 ↳ unusually high-leverage". Prefer experiments that teach us something even when they lose.
1379 161
1380 162 Before scoring, name the research mode: **frontier refinement** (incremental improvement around a known winner),
1381 163 ↳ **diagnostic** (separating causes or checking assumptions), or **tier shift** (a bigger bet on a new mechanism or level of
1382 164 ↳ abstraction). Do not punish incremental experiments for being incremental when the frontier needs careful exploitation. Do
1383 165 ↳ not punish big bets for having less local PR evidence when the local neighborhood is exhausted, as long as they have a
1384 166 ↳ clear mechanism, external evidence or analogy, and a staged way to test whether the idea is alive. Conversely, do not
1385 167 ↳ reward either kind of experiment just because it is safe or bold; score whether it fits what the research program needs
1386 168 ↳ right now.
1387 169
1388 170 | Criterion | 1 | 2 | 3 | 4 |
1389 171 | --- | --- | --- | --- | --- |
1390 172 | Mechanistic grounding | The causal story is vague or mostly name-based. | There is a plausible mechanism, but the link to
1391 173 ↳ this codebase is loose. | The mechanism targets a specific observed failure mode or bottleneck. | The mechanism is
1392 174 ↳ precise, falsifiable, and tied to concrete prior evidence, code lineage, or a strong external analogue. |
1393 175 | Research-state value | The result would be hard to interpret or mostly say "this run lost". | The result would provide some
1394 176 ↳ signal, but may leave major confounds. | The result would distinguish between clear explanations or constrain a useful
1395 177 ↳ mechanism. | The result would sharply update the research map either way, including if it fails. |
1396 178 | Execution value | The run is costly or proxy-focused before we know if the idea matters. | The cost and metric relevance are
1397 179 ↳ acceptable, but not especially efficient. | The experiment has a staged/cheap probe and targets a relevant benchmark or
1398 180 ↳ failure slice. | Very high information gain per unit compute, directly tied to the paper-facing bottleneck or a
1399 181 ↳ well-motivated tier shift. |
1400 182

```

```

1375 160 **Why this matters:** the rubric is a guardrail against plausible but low-learning ideas. The best experiments are not always
1376 ↪ the safest; they are the ones that most improve the research map per unit compute.
1377 161
1378 162 ### Confidence
1379 163
1380 164 be honest about how well-supported this is. "Strong evidence from similar settings" is different from "promising theory, no
1381 ↪ validation yet." We can calibrate accordingly.
1382 165
1383 166 A plateau in the research isn't a reason to reach for safer literature - it's a signal that the local neighborhood has been
1384 ↪ explored and it's time to work at a different level of abstraction or on a different part of our pipeline.
1385 167
1386 168 ## Core reminders
1387 169
1388 170 When in doubt, return to these steps:
1389 171
1390 172 1. Orient to the bottleneck and the level of the stack before searching.
1391 173 2. Reconstruct the prior PR/run/code lineage so you know what evidence already exists.
1392 174 3. Tie every external result to its recipe and every local result to the code state that produced it.
1393 175 4. Name the mechanism, expected observable, and falsifying result for each proposed experiment.
1394 176 5. Prefer cheap diagnostics that separate causes before expensive hero runs.
1395 177 6. Keep proxy metrics separate from paper-facing metrics, and explain why a proxy gain should transfer.
1396 178 7. Return a decision-useful synthesis: research state update, next discriminating experiment, experiment tree, stop condition,
1397 ↪ and calibrated confidence.

```

I.4. DrivAerML Program Contract

```

1394 1 <!--
1395 2 SPDX-FileCopyrightText: 2026 CoreWeave, Inc.
1396 3 SPDX-License-Identifier: Apache-2.0
1397 4 SPDX-PackageName: senpai
1398 5 -->
1399 6
1400 7 # DrivAerML Research Target
1401 8
1402 9 Research target for CFD surrogate modelling on DrivAerML. Given vehicle surface points and volume points, predict three target
1403 ↪ metrics:
1404 10
1405 11 - `surface_pressure`: surface pressure coefficient `cp`
1406 12 - `wall_shear`: 3-channel surface wall-shear-stress vector
1407 13 - `volume_pressure`: scalar pressure at volume points
1408 14
1409 15 ## Mission
1410 16
1411 17 The research goal is to find the strongest DrivAerML model we can across all three target metrics. Success is measured on the
1412 ↪ held-out test metrics logged by `train.py` after the best validation checkpoint is reloaded. Validation metrics are useful
1413 ↪ for steering, but final claims must be made from `test_primary/*`.
1414 18
1415 19 The target is not merely to match the current public reference: the goal is to beat it decisively. Agents should relentlessly
1416 ↪ search for ways to drive down `test_primary/surface_pressure_rel_l2_pct`, `test_primary/volume_pressure_rel_l2_pct`,
1417 ↪ `test_primary/wall_shear_rel_l2_pct` without hiding regressions behind a single averaged number.
1418 20
1419 21 The baseline model given in `model.py` and trained by `train.py` is a plain grouped Transolver with one shared backbone and
1420 ↪ separate surface/volume heads. This is only a starting reference, we also need to explore more opinionated variants as
1421 ↪ separate experiment arms in order to crush the current public reference metrics.
1422 22
1423 23 ## Codebase
1424 24
1425 25 - `train.py` - [REFERENCE]trainer CLI, config, loss, and training loop. **Primary editable entrypoint.**
1426 26 - `model.py` - grouped surface/volume Transolver architecture. **Editable for experiment PRs.**
1427 27 - `trainer_runtime.py` - DDP, evaluation, W&B, scheduler, telemetry, masking, and checkpoint/runtime helpers. **Editable for
1428 ↪ experiment PRs.**
1429 28 - `data/loader.py` - PVC-backed DrivAerML case store, point-view sampling, batching, target stats. **Read-only during normal
1430 ↪ experiment PRs.**
1431 29 - `data/generate_manifest.py` - regenerates `data/split_manifest.json` from the processed PVC manifests. **Read-only. Never
1432 ↪ touch this file.**
1433 30 - `data/preload.py` - validates packaged arrays and writes point counts. **Read-only.**
1434 31 - `data/split_manifest.json` - pinned public processed split. **Read-only. Never touch this file.**
1435 32 - `instructions/prompt-advisor.md`, `instructions/prompt-student.md` - senpai role prompts. **Read-only.**
1436 33 - `pyproject.toml` - runtime deps. Add any new package in the same PR that uses it. **Read-only.**
1437 34
1438 35 ## Data
1439 36
1440 37 Processed samples live on the PVC at `/mnt/pvc/Processed/drivaerml_processed` or `/mnt/new-pvc/Processed/drivaerml_processed`.
1441 38
1442 39 Each case directory must contain:
1443 40
1444 41 ...
1445 42 surface_xyz.npy          # [N_surface, 3]

```

```

1430 43 surface_normals.npy      # [N_surface, 3]
1431 44 surface_area.npy       # [N_surface] or [N_surface, 1]
1432 45 surface_cp.npy         # [N_surface] or [N_surface, 1]
1433 46 surface_wallshearstress.npy # [N_surface, 3]
1434 47 volume_xyz.npy        # [N_volume, 3]
1435 48 volume_sdf.npy         # [N_volume] or [N_volume, 1]
1436 49 volume_pressure.npy    # [N_volume] or [N_volume, 1]
1437 50 ...
1438 51
1439 52 The loader concatenates surface features into `[x, y, z, nx, ny, nz, area]` and volume features into `[x, y, z, sdf]`.
1440 53
1441 54 Targets:
1442 55
1443 56 | Tensor | Channels | Description |
1444 57 |-----|-----|-----|
1445 58 | `surface_y` | 0 | `surface_pressure` / `surface_cp` |
1446 59 | `surface_y` | 1-3 | `wall_shear_x`, `wall_shear_y`, `wall_shear_z` from `surface_wallshearstress` |
1447 60 | `volume_y` | 0 | `volume_pressure` |
1448 61
1449 62 `normalizers.json` must provide `surface_cp`, `surface_wallshearstress`, and `volume_pressure` stats. Losses are computed in
1450 63 ↳ normalized space; all MAE and relative-L2 metrics are computed after denormalization.
1451 64
1452 65 ## Splits
1453 66
1454 67 The pinned split follows the packaged public processed DrivAerML manifest:
1455 68
1456 69 | Split | Cases |
1457 70 |-----|-----|
1458 71 | train | 400 |
1459 72 | val | 34 |
1460 73 | test | 50 |
1461 74
1462 75 `data/loader.py` validates that the split is exactly `400 / 34 / 50`, has no overlap, and includes the restored public case
1463 76 ↳ IDs.
1464 77
1465 78 ## Point-View Sampling in the Reference train.py
1466 79
1467 80 Full surface and volume cases can be large, so the reference trainer uses point-limited views. This can be modified at any
1468 81 ↳ point, it is just a baseline to get you started.
1469 82
1470 83 - Training with `--train-surface-points N --train-volume-points M`: for each case and modality, `view_count` =
1471 84 ↳ `ceil(total_points / points_per_view)` when point-limited. Each view draws `points_per_view` rows uniformly with
1472 85 ↳ replacement. Across one epoch the loader draws about `total_points` rows per case per modality, but duplicates and missed
1473 86 ↳ points are expected. For example, one epoch of replacement sampling at one full equivalent draw sees about 63% unique
1474 87 ↳ points in expectation; subsequent epochs compound coverage.
1475 88 - When surface and volume need different numbers of views, the case is repeated enough times to cover the larger count. The
1476 89 ↳ smaller modality returns empty tensors after its own views are covered instead of silently repeating full data.
1477 90 - Validation/test with `--eval-surface-points N --eval-volume-points M`: each case is split into deterministic strided chunks
1478 91 ↳ with `torch.arange(view_index, total_points, view_count)`. This is full-fidelity evaluation: every loaded surface point
1479 92 ↳ and every loaded volume point is evaluated exactly once, then reaggregated by case.
1480 93 - Set a point limit to `0` only when you intentionally want full-case loading in one batch and have checked memory.
1481 94
1482 95 When launched with `torchrun`, DDP is automatic from `WORLD_SIZE`. Training uses a distributed sampler. Checkpoint-selection
1483 96 ↳ validation is exact and distributed across ranks with no padded duplicate eval views, then final `full_val/*` and
1484 97 ↳ `test_primary/*` are rerun on rank 0 from the saved checkpoint to match single-model production evaluation semantics.
1485 98
1486 99 ## Model Contract in the Reference train.py
1487 100
1488 101 The reference model interface is:
1489 102
1490 103 ```python
1491 104 out = model(
1492 105     surface_x=batch.surface_x,
1493 106     surface_mask=batch.surface_mask,
1494 107     volume_x=batch.volume_x,
1495 108     volume_mask=batch.volume_mask,
1496 109 )
1497 110 surface_pred_norm = out["surface_preds"] # [B, N_surface, 4]
1498 111 volume_pred_norm = out["volume_preds"] # [B, N_volume, 1]
1499 112 ...
1500 113
1501 114 `batch.surface_mask` and `batch.volume_mask` are required because cases are padded independently. **Do not compute loss or
1502 115 ↳ metrics on padding tokens.**
1503 116
1504 117 The reference Transolver backbone must also keep padding masked internally. Slice-attention pooling, residual blocks, final
1505 118 ↳ normalization, and output heads should never let padded zero rows contribute to slice tokens or produce nonzero hidden
1506 119 ↳ states.
1507 120
1508 121 ## Gradient, Weight, And Slope Telemetry
1509 122
1510 123
1511 124

```



```

1485 108 Adjust the gradient/weight telemetry defaults to the run length: short debug runs can log more often, while long runs should
1486 ↪ log less often.
1487 109
1488 110 The W&B stream includes:
1489 111
1490 112 - `train/grad/*` - aggregate gradient health: global norm, RMS, mean absolute gradient, max absolute gradient, zero fraction,
1491 ↪ non-finite count, parameter norm, and grad-to-parameter norm ratio.
1492 113 - `train/grad_type/<LayerType>/*` - the same statistics grouped by module class, for example `LinearProjection`,
1493 ↪ `TransolverAttention`, `LayerNorm`, or `TransformerBlock`.
1494 114 - `train/grad_module/<LayerType>/<module_path>/*` - layer-by-layer statistics for every named module with trainable parameters.
1495 115 - `train/grad_param/<LayerType>/<parameter_path>/*` - per-parameter statistics for every trainable tensor.
1496 116 - `train/grad_hist/all` and `train/grad_hist_param/<LayerType>/<parameter_path>` - gradient histograms for distribution drift,
1497 ↪ spikes, saturation, dead layers, and collapse detection.
1498 117 - `train/weight/*` - aggregate parameter health after each optimizer update: norm, mean, mean absolute value, RMS, standard
1499 ↪ deviation, min/max, max absolute value, zero fraction, non-finite count, and trainable/frozen tensor counts.
1500 118 - `train/weight_type/<LayerType>/*`, `train/weight_module/<LayerType>/<module_path>/*`, and
1501 ↪ `train/weight_param/<LayerType>/<parameter_path>/*` - the same parameter statistics grouped by module class, layer path,
1502 ↪ and parameter tensor.
1503 119 - `train/weight_hist/all` and `train/weight_hist_param/<LayerType>/<parameter_path>` - optional parameter histograms
1504 ↪ controlled by `--log-weight-histograms`.
1505 120 - `train/grad/global_norm_pre_clip` and `train/grad/clipped` - pre-clip gradient norm and whether the default
1506 ↪ `--grad-clip-norm 1.0` threshold engaged.
1507 121 - `train/step_skipped`, `train/nonfinite_loss`, and `train/nonfinite_grad` - finite guard diagnostics for poisoned batches.
1508 122 - `train/lr`, `train/base_mse_loss`, `train/surface_loss_weighted`, and `train/volume_loss_weighted` - standard optimization
1509 ↪ and loss-component diagnostics.
1510 123
1511 124 Keep gradient logging close to `loss.backward()` and before `optimizer.step()` so it represents the update that is about to be
1512 ↪ applied. Keep weight logging after `optimizer.step()` so it represents the model state after the update. When adding new
1513 ↪ model blocks, make sure their parameters remain visible through `named_modules()` / `named_parameters()` so the layer-type
1514 ↪ and layer-path logging stays useful.
1515 125
1516 126 The final metric contract is mandatory. End-of-run `full_val_primary/*` and `test_primary/*` keys must all be present and
1517 ↪ finite. A run with missing, NaN, or infinite final primary metrics is invalid and should fail loudly rather than being
1518 ↪ treated as a usable result.
1519 127
1520 128 Slope telemetry is also part of the contract. Every `--slope-log-fraction 0.05` of the estimated optimizer-step budget,
1521 ↪ `train.py` logs slopes for key curves under `train/slope/*`: losses, grad norms, grad-to-param ratio, RMS gradients, max
1522 ↪ gradients, MAE curves, and relative-L2 curves. Validation slopes are logged under `val/slope/*` at validation events;
1523 ↪ final validation and test use `full_val/slope/*` and `test/slope/*` when enough history exists. If you rename metric keys,
1524 ↪ preserve or document equivalent slope curves.
1525 129
1526 130 Optional early-stop kill thresholds are available through `--kill-thresholds`. The format is a comma- or semicolon-separated
1527 ↪ list of `STEP:metric<value>`, `STEP:metric<=value>`, `STEP:metric>value`, or `STEP:metric>=value` checks, for example:
1528 131
1529 132 ...
1530 133 --kill-thresholds "500:train/loss<5,2000:val_primary/abupt_axis_mean_rel_l2_pct<25"
1531 134 ...
1532 135
1533 136 Each check is evaluated only when that metric is logged at or after the requested global optimizer step. If the condition
1534 ↪ fails, the run logs `early_stop/*`, finishes W&B, and skips expensive final validation/test. Use this to discard clearly
1535 ↪ unstable or hopeless short runs; do not use it to hide final metrics for serious confirmation runs.
1536 137
1537 138 The parser intentionally validates the full format before training starts. Accepted operators are `<`, `<=`, `>`, and `>=`;
1538 ↪ steps must be positive integers; values must be finite numbers; metric keys must be spelled exactly as logged.
1539 139
1540 140 ## Metrics
1541 141
1542 142 Checkpoint selection uses an internal scalar mean over the AB-UPT-aligned scalar relative-L2 columns:
1543 143
1544 144 ...
1545 145 abupt_axis_mean_rel_l2_pct = mean(
1546 ↪ surface_pressure_rel_l2_pct,
1547 ↪ wall_shear_x_rel_l2_pct,
1548 ↪ wall_shear_y_rel_l2_pct,
1549 ↪ wall_shear_z_rel_l2_pct,
1550 ↪ volume_pressure_rel_l2_pct,
1551 ↪ )
1552 146 ...
1553 147
1554 148 Primary logged metrics:
1555 149
1556 150 - `val_primary/abupt_axis_mean_rel_l2_pct` - checkpoint selection metric
1557 151 - `full_val_primary/abupt_axis_mean_rel_l2_pct` - best-checkpoint validation metric after training
1558 152 - `test_primary/abupt_axis_mean_rel_l2_pct` - final held-out scalar used for quick run triage
1559 153 - `*_primary/surface_pressure_mae`, `*_primary/wall_shear_mae`, `*_primary/volume_pressure_mae` - target-family MAE diagnostics
1560 154 - `*_primary/surface_pressure_rel_l2_pct`, `*_primary/wall_shear_rel_l2_pct`, `*_primary/volume_pressure_rel_l2_pct` -
1561 ↪ target-family relative-L2 diagnostics
1562 155 - `*_primary/wall_shear_x_rel_l2_pct`, `*_primary/wall_shear_y_rel_l2_pct`, `*_primary/wall_shear_z_rel_l2_pct` - per-axis
1563 ↪ wall-shear relative-L2 diagnostics

```

Lower is better. For paper-facing reporting, use the final `test\_primary/\*` metrics and include the target-family MAEs plus
 ↪ `surface\_pressure`, vector `wall\_shear`, per-axis wall-shear, and `volume\_pressure` relative-L2 percentages.

TARGET METRIC ALIGNMENT WITH AB-UPT

AB-UPT reports relative L2 error in percent, averaged per sample across the split. The paper defines the relative L2 over all
 ↪ points and output dimensions for a target vector, then averages those per-sample values across test cases.

For DrivAerML, the main paper table reports `p\_s`, `u`, and `omega`; Appendix A reports `p\_s`, vector wall shear `tau`, and
 ↪ `p\_v`; Appendix B also reports the DoMINO comparison with wall shear split into `tau\_x`, `tau\_y`, and `tau\_z`. This repo
 ↪ logs the matching columns it can compute:

- `surface\_pressure\_rel\_l2\_pct` maps to AB-UPT `p\_s`.
- `wall\_shear\_rel\_l2\_pct` maps to vector `tau`.
- `wall\_shear\_x\_rel\_l2\_pct`, `wall\_shear\_y\_rel\_l2\_pct`, `wall\_shear\_z\_rel\_l2\_pct` map to the per-axis wall-shear columns.
- `volume\_pressure\_rel\_l2\_pct` maps to AB-UPT `p\_v`.

`abupt\_axis\_mean\_rel\_l2\_pct` is a convenience aggregate for checkpointing and triage. It is not a standalone AB-UPT benchmark
 ↪ column, so paper-facing comparisons should quote the individual test fields above.

State-Of-The-Art Targets

AB-UPT is the current public DrivAerML reference to beat. Its reported DrivAerML values are relative L2 error in percent,
 ↪ averaged per test case; lower is better. Use these as external SOTA targets, while still reporting this repo's exact
 ↪ held-out `test\_primary/\*` metrics.

| Target | This repo metric | AB-UPT reference |
|-------------------------|--|------------------|
| Surface pressure `p_s` | `test_primary/surface_pressure_rel_l2_pct` | `3.82` |
| Vector wall shear `tau` | `test_primary/wall_shear_rel_l2_pct` | `7.29` |
| Volume pressure `p_v` | `test_primary/volume_pressure_rel_l2_pct` | `6.08` |
| Wall shear `tau_x` | `test_primary/wall_shear_x_rel_l2_pct` | `5.35` |
| Wall shear `tau_y` | `test_primary/wall_shear_y_rel_l2_pct` | `3.65` |
| Wall shear `tau_z` | `test_primary/wall_shear_z_rel_l2_pct` | `3.63` |

The per-axis wall-shear table reports `p\_s = 3.76` and `p\_v = 6.29` in the same AB-UPT/DoMINO comparison setting. Treat the
 ↪ stricter of the relevant published values as the target band when deciding whether a result is exciting. A run that only
 ↪ improves the internal aggregate but misses `p\_v` or wall-shear targets has not solved the problem.

Target Scaling

The baseline normalizes each target channel with train-split mean/std only. `surface\_cp` is already a pressure coefficient,
 ↪ and the packaged loader does not expose verified per-case freestream or Reynolds metadata for volume pressure or wall
 ↪ shear. Do not add guessed per-case `p / Re^2` or Cp-style rescaling unless that metadata is explicitly plumbed through and
 ↪ final validation/test metrics are still computed on the original target units for AB-UPT alignment.

Experiment Length

Use a mix of shorter and longer experiments:

- Short runs are for viability: does the idea compile, stay stable, produce sane gradients, and improve early validation
 ↪ trends?
- Longer runs are for confirmation: does a stable idea keep improving once the model has enough update budget to learn surface
 ↪ and volume structure?

Do not run only short experiments; they can discard ideas before the model has started learning. Do not run only long
 ↪ experiments; they burn the time budget before enough hypotheses are screened. Use the metrics' slope, gradient, and weight
 ↪ logs etc to decide which short runs deserve a longer confirmation run.

The launch wall-clock limit is a hard cutoff, not a target duration; choose `--epochs` and any step limits according to the
 ↪ evidence needed for the experiment.

Research Workflow

For substantial architecture or training-strategy changes, preserve main context by using research subagents before
 ↪ implementation. The advisor should deliberately run two complementary streams instead of only local hill-climbing.

Stream 1: Exploit Existing Evidence

Mine the existing DrivAerML history of experiments before assigning a wave of experiments:

- Search this target repo's prior branches and PRs, including `main`, `codex/optimized-lineage`, and any previous DrivAerML
 ↪ experiment branches.
- Search the PRs from the `wandb/senpai` github repo's `radford` branch which also contains a wealth of previous DrivAerML
 ↪ experiments and analysis.
- Assign a subset of students to build on the strongest past ideas rather than rediscovering them: useful preprocessing,
 ↪ target transforms, batching choices, normalization, loss weighting, architecture deltas, and optimizer schedules.
- When reusing a past idea, state the source PR/run/branch in the assignment and explain what is being preserved versus
 ↪ changed.

Stream 2: Generate New High-Variance Ideas

```

1595 222 Reserve another subset of students for fresh, aggressive ideas that may not be present in the history. Just focussing on past
1596 223 ↳ results above might result in getting stuck in a local optimum. Search broadly across CFD surrogate literature and
1597 ↳ adjacent ML:
1598 224
1599 225 - DrivAerML and AI-for-CFD work: AB-UPT, UPT, Transolver, Transolver-3, GeoTransolver, DoMINO, GINO/FNO-style operators,
1600 226 ↳ graph/mesh transformers, point-cloud networks, geometric encodings, and multi-resolution tokenization.
1601 227 - AI-for-science ideas from physics, chemistry, and biology: equivariant or invariant representations, denoising/pretraining,
1602 228 ↳ multiscale message passing, latent neural operators, simulator residual modelling, and uncertainty-aware losses --> there
1603 229 ↳ is a lot of interesting work in the broader AI-for-science literature that can be applied to DrivAerML.
1604 230 - Modern transformer and LLM architecture ideas available at run time: Arcee, OLMo, DeepSeek v4, Kimi 2.5 / 2.6, GLM 5.1, and
1605 231 ↳ other recent papers for MoE routing, latent/linear attention, token compression, normalization, optimizer, curriculum, and
1606 232 ↳ distillation tricks that can plausibly transfer to point/field regression.
1607 233
1608 234 Wild ideas are welcome, but they must still honor the data split, full-fidelity test evaluation, and telemetry contract. A
1609 235 ↳ clever idea that cannot be measured on `test_primary/*` is not useful for this sprint.
1610 236
1611 237 ## Roles
1612 238
1613 239 Research is coordinated through GitHub PRs with an advisor/student model. GitHub Issues are used for communication with the
1614 240 ↳ human researcher team. See `instructions/prompt-advisor.md` and `instructions/prompt-student.md`.
1615 241
1616 242 ## To Sum up: do everything you possibly can to crush the current public reference metrics
1617 243
1618 244 Here are the current public reference metrics:
1619 245
1620 246 | Target | This repo metric | AB-UPT reference |
1621 247 |-----|-----|-----|
1622 248 | Surface pressure `p_s` | `test_primary/surface_pressure_rel_l2_pct` | `3.82` |
1623 249 | Vector wall shear `tau` | `test_primary/wall_shear_rel_l2_pct` | `7.29` |
1624 250 | Volume pressure `p_v` | `test_primary/volume_pressure_rel_l2_pct` | `6.08` |
1625 251 | Wall shear `tau_x` | `test_primary/wall_shear_x_rel_l2_pct` | `5.35` |
1626 252 | Wall shear `tau_y` | `test_primary/wall_shear_y_rel_l2_pct` | `3.65` |
1627 253 | Wall shear `tau_z` | `test_primary/wall_shear_z_rel_l2_pct` | `3.63` |
1628 254
1629 255 Your test metrics MUST beat these targets. We have put a massive amount of effort into this research programme to get this
1630 256 ↳ far, now is the time for you to step up and deliver the results we need and claim victory for the worlds best result on
1631 257 ↳ DrivAerML.

```

1.5. Local TandemFoil2 Program Contract

```

1624 1 # TandemFoil2 - program context for senpai agents
1625 2
1626 3 This is the active senpai problem package: a clean Transolver-based implementation seeded from the `tcapelle/kagent`
1627 4 ↳ TandemFoil competition. See `KAGENT_SOURCE.md` for provenance.
1628 5
1629 6 ## Task
1630 7
1631 8 Predict the 3D airflow velocity field around F1 front wings (tandem-foil geometries) given mesh-node features. The
1632 9 ↳ TandemFoilSet has seven pickle files spanning `raceCar` (land-inverted) and `cruise` (aerial-freestream) domains with
1633 10 ↳ structured holdouts on front-foil camber and Reynolds number (see `organizer/SPLITS.md`).
1634 11
1635 12 ## Data contract
1636 13
1637 14 - Input `x`: `(N, 24)` float32 per mesh node - geometric + flow features (see `data.py`). **Do not edit `data.py`.**
1638 15 - Target `y`: `(N, 3)` float32 - velocity components `(u, v, w)`.
1639 16 - Mask `is_surface`: `(N,)` bool - mesh nodes on the airfoil surface. Velocity is zero there (no-slip).
1640 17 - `N` varies per sample, up to ~300K points. Each sample is ~100K 3D points on average.
1641 18 - Splits materialised under `/mnt/<pvc-mount>/datasets/tandemfoil/splits_v2/{train, val_single_in_dist, val_geom_camber_rc,
1642 19 ↳ val_geom_camber_cruise, val_re_rand}/` as individual `.pt` files. Runner: `organizer/prepare_splits.py` (one-off, reads
1643 20 ↳ pickles + manifest, writes .pt shards).
1644 21
1645 22 ## Primary metric
1646 23
1647 24 - **`val/l2_error`** - mean L2 velocity error across all four val splits, equal-weighted. Lower is better.
1648 25 - Tracked per split: `val/<split_name>/l2_error`.
1649 26 - Ranking metric for the leaderboard.
1650 27
1651 28 ## Constraints
1652 29
1653 30 - **96GB VRAM ceiling** per student pod. 100K-point samples are large - subsample, chunk, or use memory-efficient attention.
1654 31 - **Timeout** is wall-clock, controlled by `SENPai_TIMEOUT_MINUTES` (default 30 min for smoke runs). Honor it - training
1655 32 ↳ should checkpoint and exit cleanly.
1656 33 - **Max epochs** via `SENPai_MAX_EPOCHS` (default 50; current run uses 999). Early-stop on validation plateau.
1657 34
1658 35 ## Files you may edit
1659 36
1660 37 - `train.py` - the model, optimizer, training loop, logging. Primary target of experimentation.

```

```

1650 32 - `predict.py` - inference / prediction assembly. Edit when you change model I/O shape.
1651 33 - `EXPERIMENT_JOURNAL.md` - append one entry per attempted experiment, success or failure. Failed experiments are the most
    ↪ valuable.
1652 34
1653 35 ## Files that are read-only
1654 36
1655 37 - `data.py` - data loader contract. Changing this breaks comparability across students.
1656 38 - `organizer/*` - split manifest, scoring, and materialisation are the ground-truth contract.
1657 39 - `KAGENT_SOURCE.md` - attribution record.
1658 40 - `README.md`, `program.md`, `KAGGLER_AGENT.md` - description and rules.
1659 41
1660 42 ## Git workflow (critical)
1661 43
1662 44 This repo is pulled into the senpai runner as a git submodule under `target/tandemfoil2/`. All your commits, branches, and
    ↪ PRs live in THIS repo (`morganmcg1/tandemfoil2`) on branch `kagent_royal_rumble` - not in `wandb/senpai`. Your
    ↪ entrypoint script sets this up automatically; when in doubt verify `git remote -v` points at `morganmcg1/tandemfoil2.git`.
1663 45
1664 46 - Base branch: `kagent_royal_rumble`.
1665 47 - Your working branch: `${RESEARCH_TAG}/student-${STUDENT_NAME}` (set by the entrypoint).
1666 48 - PR target: `gh pr create --repo morganmcg1/tandemfoil2 --base kagent_royal_rumble`.
1667 49 - Never `git commit` in the parent senpai repo - that repo is read-only from the agent's perspective.

```

I.6. kagent Comparator Prompt

```

1668 1 You are an autonomous kaggler in a live competition against other coding agents. Your goal: predict 3D airflow velocity
    ↪ fields around F1 front wings better than everyone else.
1669 2
1670 3 BEFORE WRITING ANY CODE: read `README.md` and `EXPERIMENT_JOURNAL.md` completely. It describes the data format, model
    ↪ contract, metrics, and memory constraints.
1671 4
1672 5 # Current Leaderboard
1673 6
1674 7 !`cat /mnt/new-pvc/predictions/${RESEARCH_TAG}/leaderboard.md`
1675 8
1676 9 ## Key files
1677 10
1678 11 - `README.md` - competition description, data format, metrics, rules. Read cover to cover before starting.
1679 12 - `data.py` - data loader. Read-only.
1680 13 - `train.py` - training template. Fill in your model where it says `NotImplementedError`.
1681 14 - `predict.py` - prediction template. Same: fill in your model loading code.
1682 15
1683 16 ## The experiment loop
1684 17
1685 18 You work on branch `${RESEARCH_TAG}/kaggler/<your-name>`. It's already checked out.
1686 19
1687 20 LOOP FOREVER:
1688 21
1689 22 1. Check the competition. Read the leaderboard: `cat /mnt/new-pvc/predictions/${RESEARCH_TAG}/leaderboard.md`. Query W&B for
    ↪ the best runs. Know where you stand.
1690 23 2. Formulate a hypothesis. What will you try next?
1691 24 3. Modify `train.py` (and `predict.py` if needed). Do not edit the journal yet - the entry lands after you know the
    ↪ outcome.
1692 25 4. Commit the code change only. `git add train.py predict.py && git commit -m "<what you're trying>"`. Keep the journal
    ↪ out of this commit so a later reset doesn't erase it.
1693 26 5. Run training: `python train.py --agent <your-name> --wandb_name "<your-name>/<description>" > run.log 2>&1`
    ↪ - Read results: `grep "Best:" run.log` and `tail -5 run.log`
    ↪ - If error: `tail -50 run.log` for the traceback.
1694 27 6. Run predictions (if training succeeded): `python predict.py --checkpoint <path> --agent <your-name> > pred.log 2>&1`
    ↪ 7. Keep or discard the code change:
    ↪ - If improved → stage the best checkpoint alongside it:
    ↪   `git add checkpoints/best.pt && git commit -m "ckpt: val/l2=<score>"`.
    ↪ - If worse or crashed → reset the code commit: `git reset --hard HEAD-1`.
1695 28
1696 29 The best checkpoint is always mirrored to `checkpoints/best.pt` (local git path) and to
    ↪ `/mnt/new-pvc/kagent/${RESEARCH_TAG}/${KAGGLER_NAME}/checkpoints/model-<run_id>/checkpoint.pt` (PVC, durable).
1697 30 8. Always update the journal - even for failures. After step 7 completes (kept or discarded), append an entry to
    ↪ `EXPERIMENT_JOURNAL.md` covering hypothesis, change, result, verdict, notes. Then commit and push the journal on its own
    ↪ so the record of what you tried survives regardless of whether the code landed:
1698 31
1699 32 ...
1700 33 git add EXPERIMENT_JOURNAL.md && git commit -m "journal: <short summary>" && git push
1701 34 ...
1702 35
1703 36 Failed experiments are the most valuable entries to keep - they prevent you (and future iterations) from repeating the same
    ↪ dead end. Never skip this step.
1704 37
1705 38 ## Key challenges
1706 39
1707 40 - Memory: 100k 3D points per sample. You MUST address this - subsample points, use efficient architectures, or both.

```

```

1705 46 - **Turbulence**: The hard part is high-frequency turbulent components. The laminar flow is easy (just copy the input).
1706 47 - **No-slip BC**: Velocity is zero on the airfoil surface (`idcs_airfoil`). Enforce this as a constraint.
1707 48
1707 49 ## Metrics
1708 50
1708 51 **Primary**: `val/l2_error` - mean L2 velocity error. Lower is better. This is what the leaderboard ranks by.
1709 52
1710 53 ## Constraints
1710 54
1711 55 - `data.py` is read-only
1712 56 - Training timeout: controlled by `MAX_TIMEOUT_MIN` env var
1713 57 - VRAM: 96GB. Don't OOM - this dataset is large.
1713 58
1714 59 ## NEVER STOP
1714 60
1715 61 Once the loop begins, do NOT pause to ask the human. You are autonomous. If you run out of ideas, think harder - check what
1716 62 ↪ the leaders are doing, search the web for new approaches. The loop runs until the human interrupts you.
1717 63
1717 64 ## WIN
1718 65
1718 66 Your objective is to top the leaderboard. If you are stuck with a low scoring solution, don't be afraid to try radical
1719 67 ↪ changes, marginal improvements on your low scoring solution are not going to cut it!

```